

Internet

A



B

Data Communications & Computer Networks

Tutorial for Exit Exam

BSc In IT,CS and SE

Themes	Course Name	Credit hour	Weight of course or proportion	Number of test items from each course	Cognitive						Affective	Psychomotor	Total
					Remember	Understand	Apply	Analyze	Evaluate	Create			
	Total credit	12											
Computer Networking and Security Share = 18 items from the total	Data Communication and Computer Networking	3	3/9 = 0.33	0.33*18 = 6	-	3	1	1	1	-	-	-	6
	Computer Security	3	3/9 = 0.33	0.33*18 = 6	-	2	3	1	-	-	-	-	6
	Network and System Administration	3	3/9 = 0.33	0.33*18 = 6	-	1	1	2	1	1	-	-	6
	Total credit	9											

Table 6-2: Mapping Learning Outcomes to Skill Sets

Themes	Course Name	General Objective	Learning outcomes	Cognitive						Affective	Psychomotor	Total
				Remember	Understand	Apply	Analyze	Evaluate	Create			
CS Exit Exam Blueprint for Data Communication												
Computer Networking and Security	Data Communication and Computer Networking	Understand the fundamentals of data communications networks, including protocol operation and the concepts and operation of local	Understand the concepts and principles of data communications and computer networks	-	1	-	-	-	-	-	-	1
			Understand Protocols and various networking components	-	1	-	-	-	-	-	-	1
	Number of test items from the course =6	and IP-based networks.	Understand TCP/IP & OSI Reference Model	-	1	-	-	-	-	-	-	1
Understand Design and implement IP addressing and subnets			-	-	1	1	1	-	-	-	3	

IT Exit Exam Blueprint for Data Communication

IT Exit Exam Blueprint for Data Communication													
Themes and Share of the themes in %	Name and Share of the Courses in %	Cred it hour	General objective/Competency	Specific objectives /learning outcomes	Learning outcomes								No. of items
					Cognitive						Affective	Psychomotor	
					Remembering	Understanding	Application	Analysis	Evaluation	Creation/Synthesi			
Networking and Information Security	Data Communication and Computer Networks	3	Understands data transmission and media, protocols, the concept of layering, recognize the importance of networking standards	Students will be able to understand data transmission and transmission media and protocols		3							3
				Students will identify the seven layers of the OSI model.				3					3
				Students will have a deep knowledge of TCP/IP and OSI model.		2							2

5.	Networking and Information Security	1. Data communications and Computer Networks	3
		2. System & Network Administration	3
		3. Network Device & Configuration	3
		4. Information Assurance and Security	2

[illegible]

SE Exit Exam Blueprint for Fundamental of Networking

Themes and Share of the themes in %	Name and Share of the Courses in %	ECTS	General objective /Competency	Specific objectives /learning outcomes	Learning outcomes								No. of items
					Cognitive						Affective	Psychomotor	
					Remembering	Understanding	Application	Analysis	Evaluation	Creation/Synthesi			
Networking and Software Security 11.49	Fundamentals of networking 0.5	5	Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.	Describe networking protocols and standards;	1								6
			Identify reference models and layered architectures in networking;	1									
			Recognize network topologies, media types and network components;		1								
			Identify the network (OSI) layers at which various network devices operate;	1									
			Apply basic cabling and network design techniques to connect devices.						2				

PART I

Basics of Data Communications

OBJECTIVES:

- ❖ Understand the concepts and principles of data communications
- ❖ Understand Data transmission and Transmissions medias

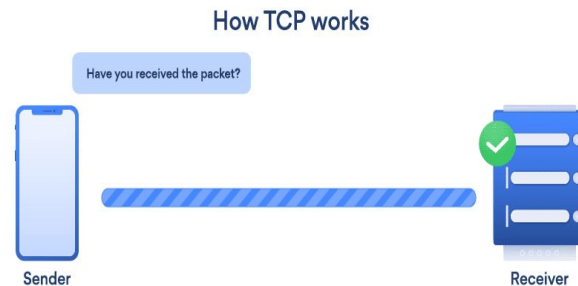
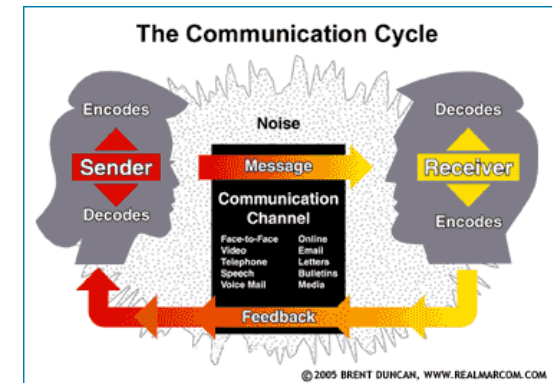
Overview Data Communications

- When we communicate, we are **sharing** information.
- This sharing can be **local** or **remote** between individuals, local communication usually occurs face to face, while remote communication takes place **over distance**.
 - *Telecommunication*
- For data communications to occur, the communicating devices must be part of a **communication system** made up of a combination of **hardware** and **software**
- The effectiveness of a data communications system depends on **four fundamental characteristics**:
 - **Delivery,**
 - **Accuracy**
 - **Timeliness and**
 - **Jitter---QOS**

What is Data Communications?

- Is the process of transferring digital information **between two or more points**.
- In Data Communications, **data** generally are defined as information that is stored in digital form
- Data communications can be summarized as the **transmission, reception, and processing** of digital information
- A data communications system has **five components**:

- **Message**
- **Sender**
- **Receiver**
- **Transmission medium**
- **Protocol**

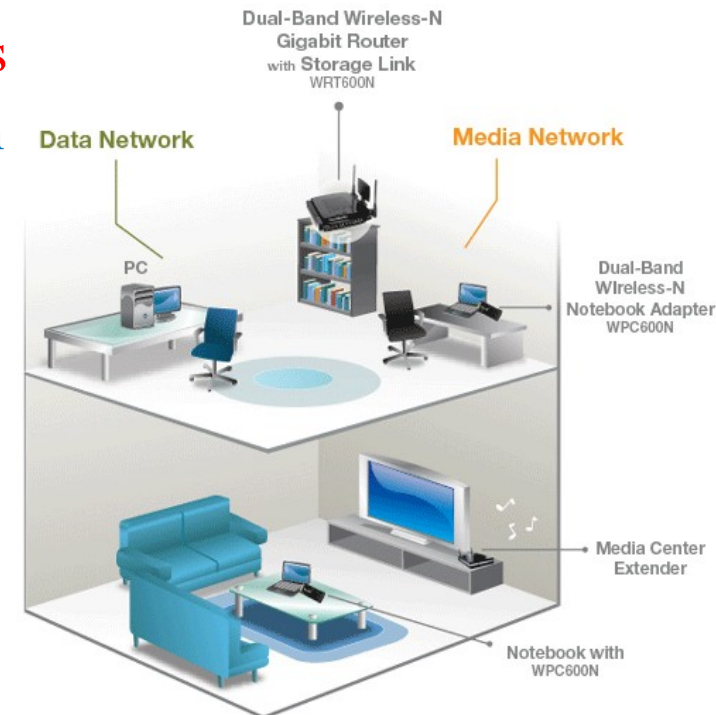
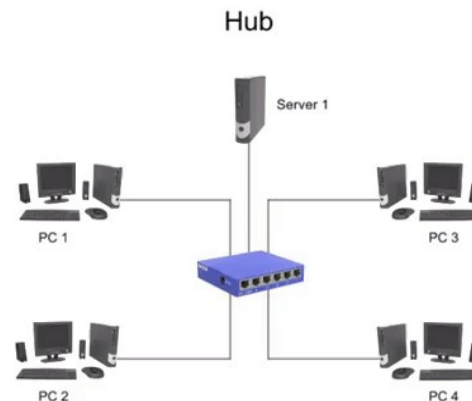
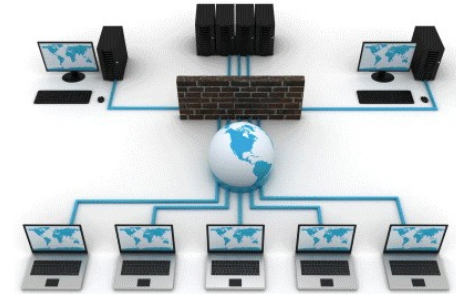


Cont...

- Information today comes **in different forms** such as
 - Text
 - Numbers
 - Images
 - Audio and
 - Video
- All of these data type has their own **representation** and **coding** system in the computer systems.

What is a computer network?

- A computer network comprises
 - ...two or more computers that are connected
 - ...either by cables (wired) or WiFi (wireless)
 - ...with the purpose of transmitting, exchanging, or sharing data and resources
- Refers to
 - ...a group of two or more computing devices
 - ... that are connected by a communication medium
 - ...allowing the computers to communicate electronically



Data Transmission

❑ It is sending of binary data from source to destination as **bit by bit** or **grouping** bits together, if so, **how?**

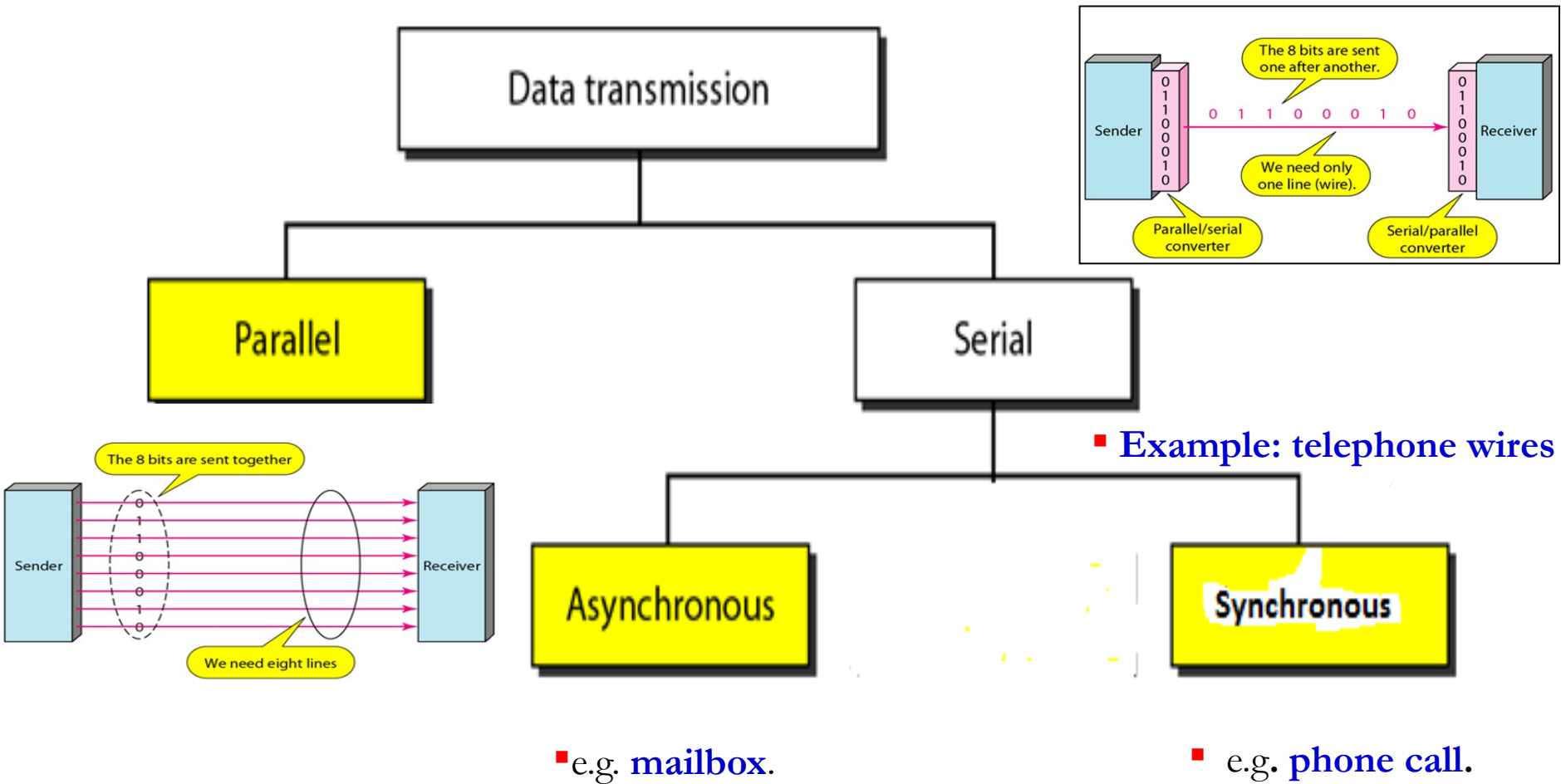
❑ **Ans:**

❖ The transmission of binary data across a link (communication channel) can be either **parallel** or **serial** mode.

❑ **This means?**

❖ **Parallel mode** **multiple** bits are sent together each clock, where as in **serial mode**, **1 bit** is sent with each clock tick.

Types/ways of Data Transmission



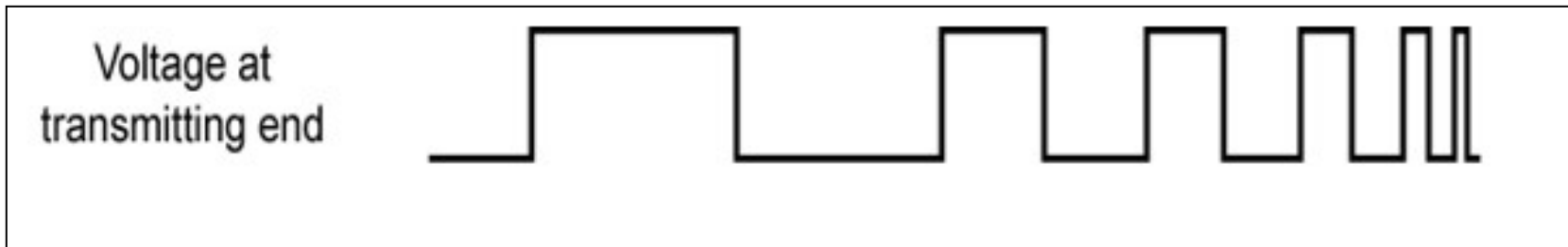
SIGNALS

Signals...

- In a **communication system**, data are propagated from one point to another by means of **electromagnetic signals**.
- Now we consider the signal generated by the **transmitter**.
 - Propagated over a variety of **media**: **wire**, **fiber optic**, **space**
- **Two type of signals**
 - **Digital** and
 - **Analog**

1. Digital Signals

- A **digital signal** is one in which the signal intensity maintains a constant level for some period of time
- Used to represent *digital data*



- **Advantages of Digital Signals**
 - Less vulnerable to noise

2. Analog Signals

- **Analog signal** is the simplest sort of signal is a in which the same signal pattern repeats over time and
- Used to represent *analog data*
 - **For example**, when you speak into a microphone, the sound waves are converted into analog electrical signals that can be transmitted or processed.

- Can use **analog signal to carry digital data ?**

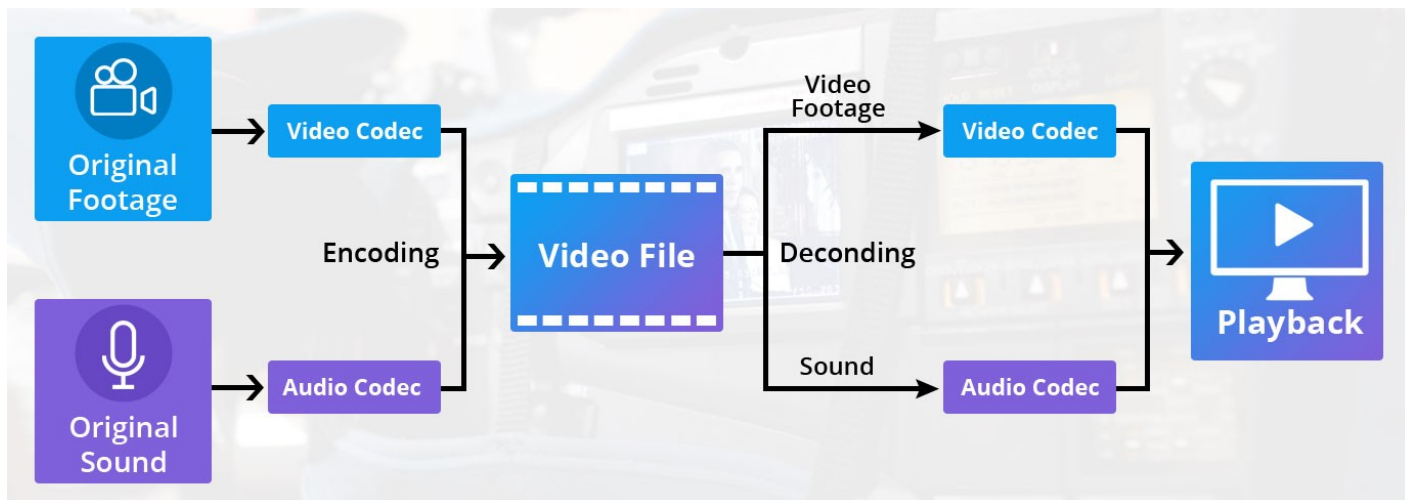
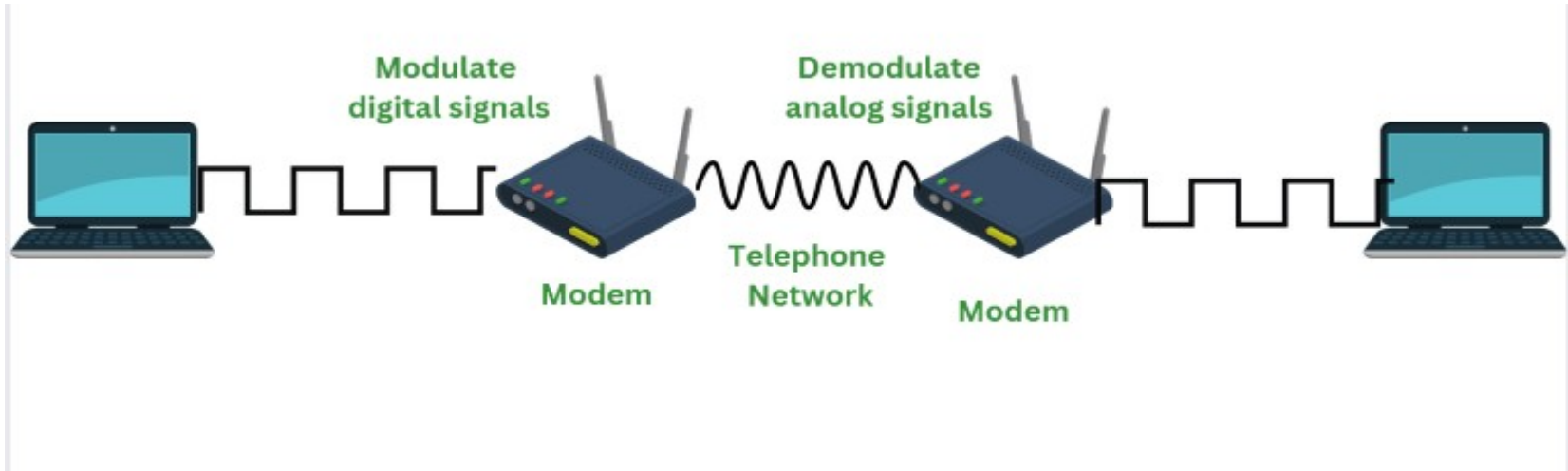
- By using?

- **Modem**: modulator/demodulator converts a series of **binary voltage pulses into an analog signal by encoding** the digital data onto a carrier frequency.

- Can use **digital signal to carry analog data?**

- **Codec** (coder-decoder): it takes an analog signal that directly represents the voice data and approximates that signal by a bit stream.

MODEM



CODEC

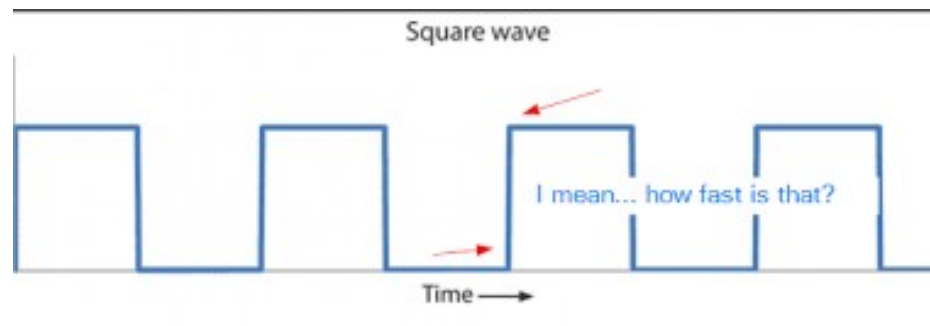
I. Digital Data Transmission

1. Digital data

- Produced by **computers**, in binary form, represented as a series of **ones** and **zeros**

2. Digital transmissions

- Made of **square waves** with a clear beginning and ending
- Computer networks send digital data using digital transmissions.



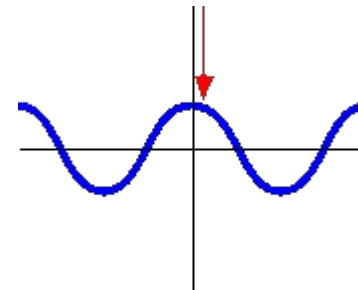
II. Analog Data Transmission

1. Analog data

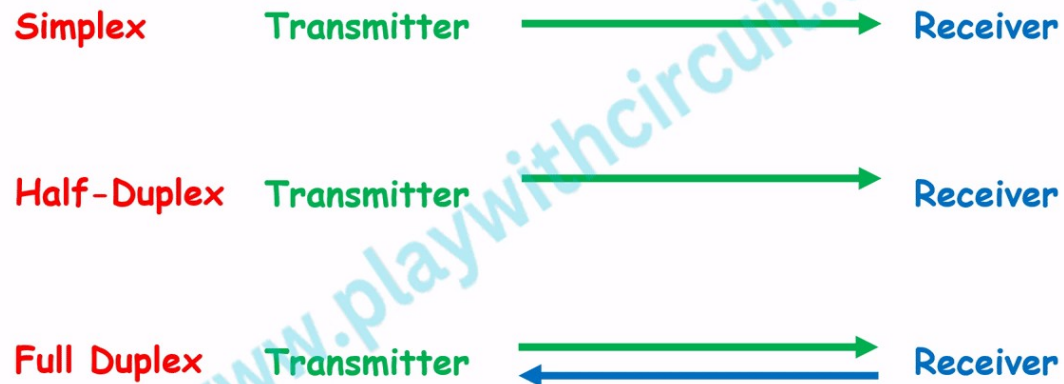
- Produced by **telephones**
- Sound waves, which vary continuously over time
- Can take on any value in a wide range of possibilities

2. Analog transmissions

- Analog data transmitted in analog form (vary continuously)
- Examples of analog data being sent using analog transmissions are broadcast **TV** and **radio**



DATA TRANSMISSION MODE

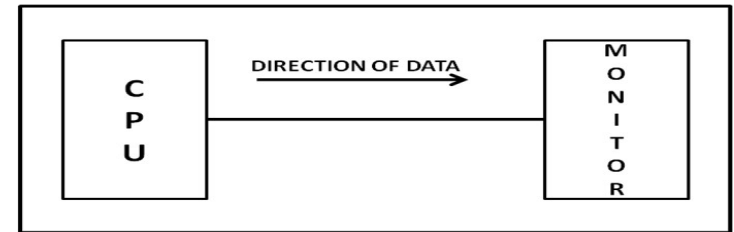


Data Transmission Mode

- It is also called **Communication Mode**.

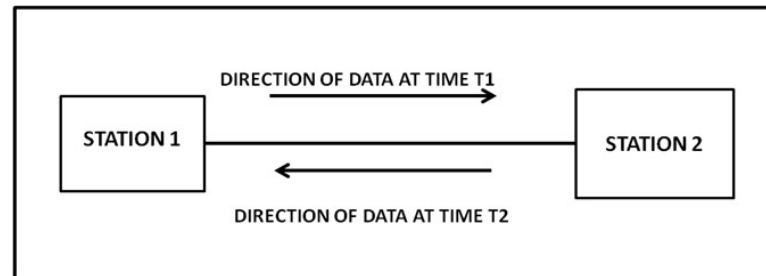
1. Simplex transmission

- Signals are transmitted in **only one(unidirectional)** direction e.g. **Television**



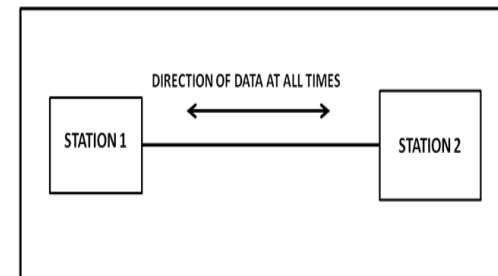
2. Half duplex

- Signals can be transmitted in **either** direction, but only one way at a time. e.g. **police radio**



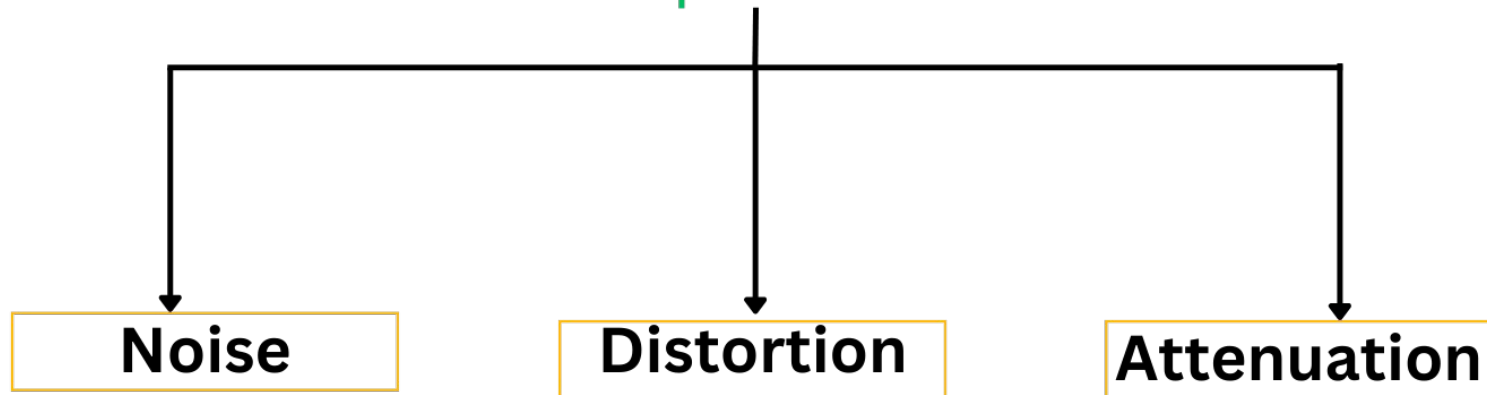
3. Full duplex

- Both stations may transmit **simultaneously**. e.g. **telephone**



TRANSMISSION IMPAIRMENTS

Transmission Impairments
In Computer Network



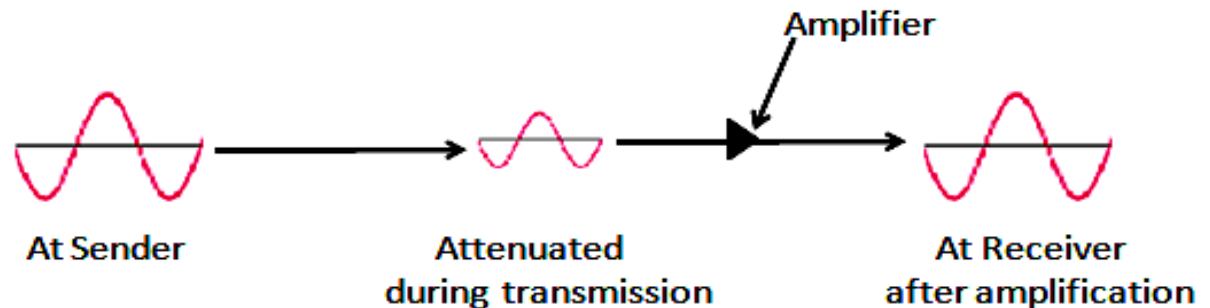
Transmission Impairments and their solution...

- **Consequences** over two types of signals we have:
 - For analog signals: degradation of signal **quality** (if distance increases)
 - For digital signals: bit **errors**
- The most significant **causes** for transmission impairments include
 - a. **Attenuation**
 - b. **Distortion**
 - c. **Noise**

Transmission Impairments and their solution...

1. Attenuation

- Attenuation means a **loss** of energy.
- It is commonly **observed in wired communication** systems, where the signal strength decreases with **increasing cable length**.
- Attenuation can also occur in **wireless communication** due to **obstacles**, such as **buildings** or environmental factors like **atmospheric conditions**.



Transmission Impairments and their solution...

2. Distortion

- Means that the signal changes its form or shape distortion occurs in composite signals

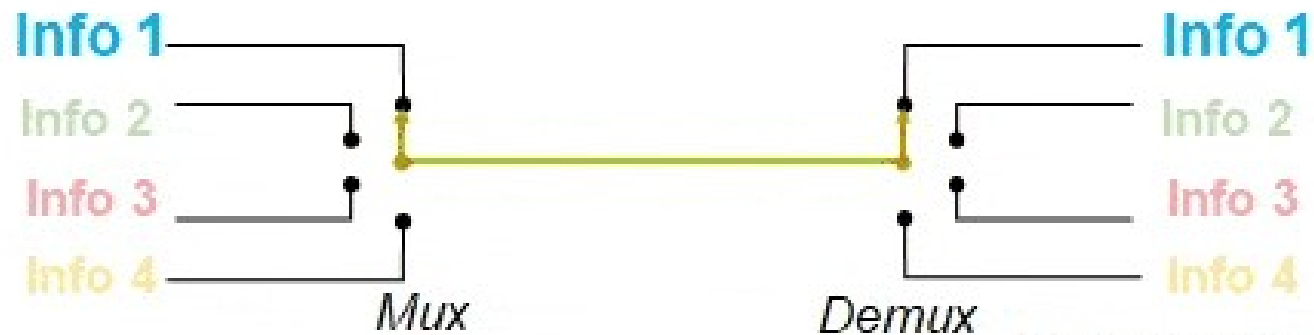


Transmission Impairments and their solution...

3. Noise:

- Refers to any **unwanted** interference that disrupts the original signal.
- It can be caused by external factors such as **electromagnetic interference (EMI)** from other electronic devices, **radio frequency interference (RFI)**, or internal factors like thermal noise within the communication system components.

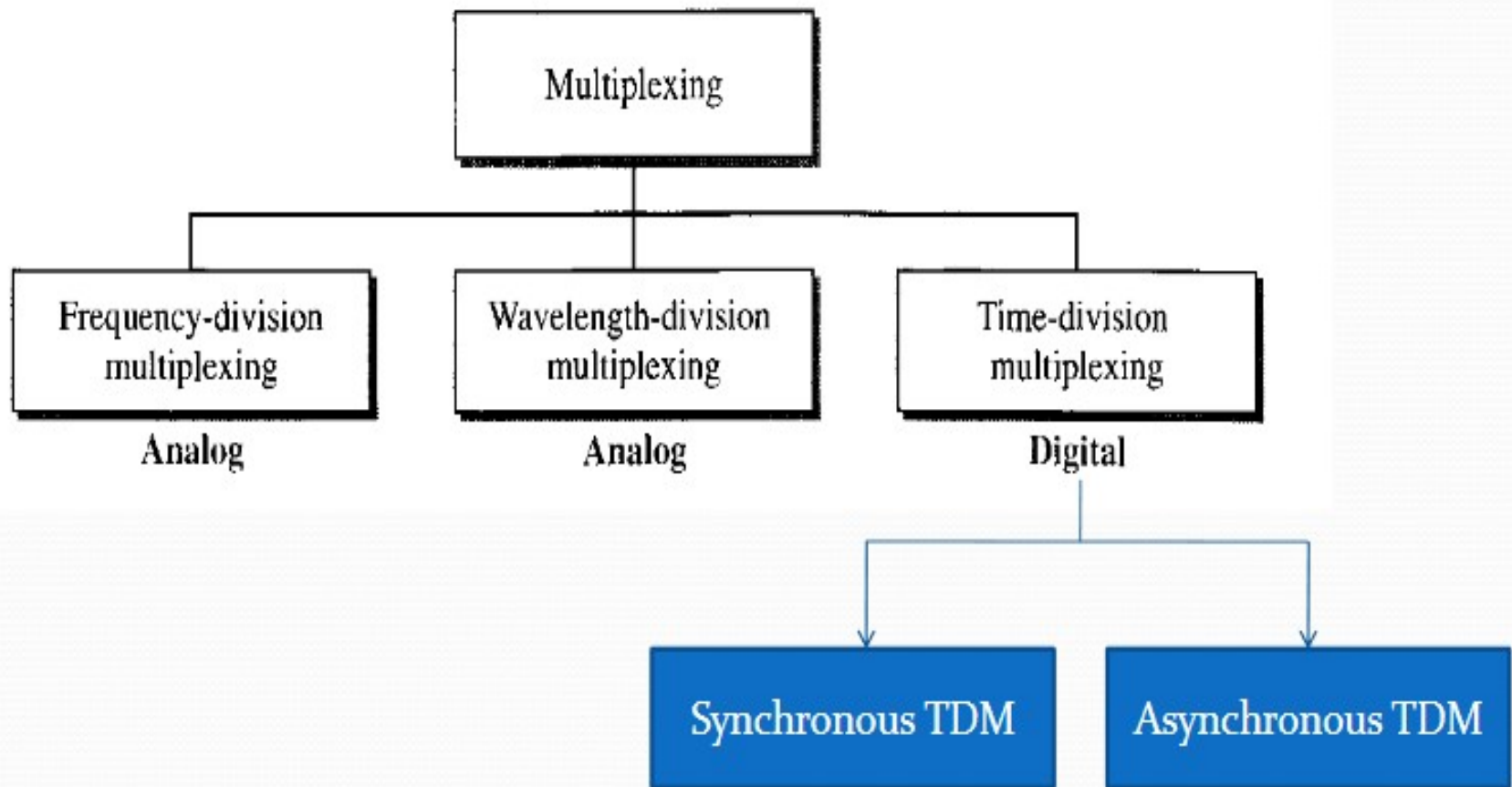
MULTIPLEXERS



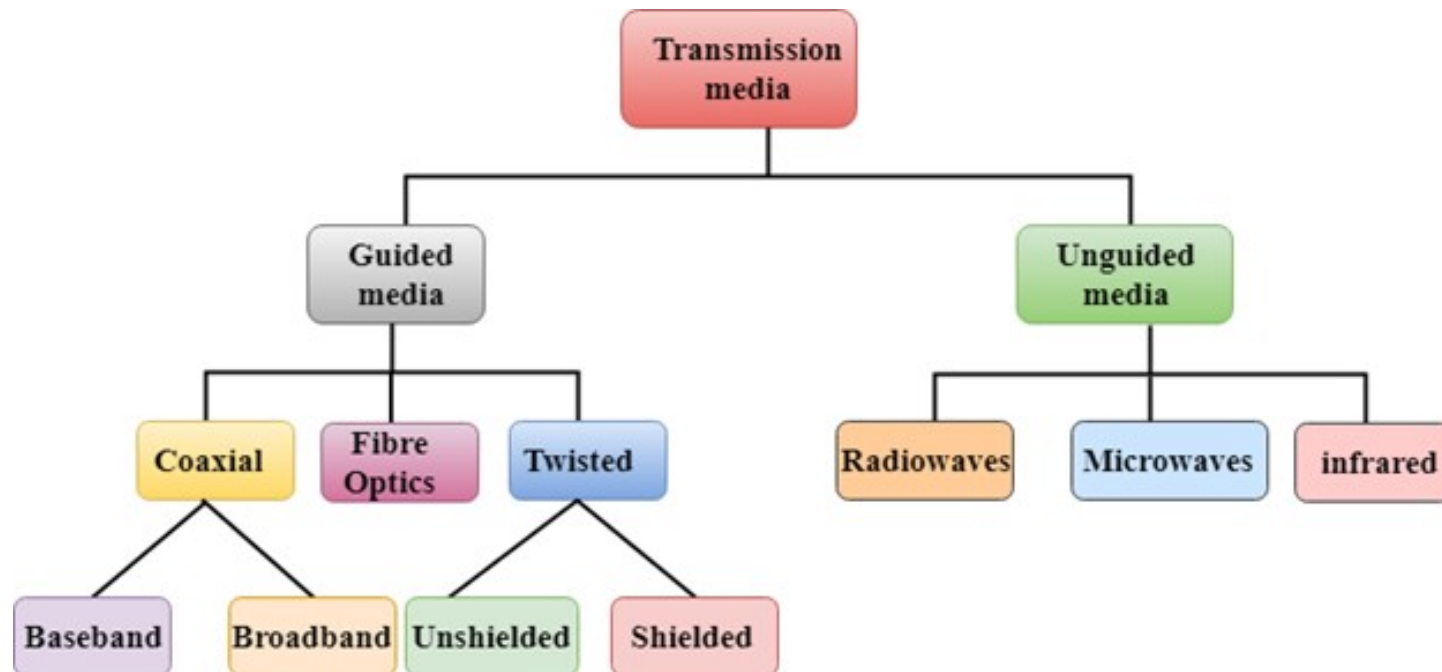
Multiplexing/Demultiplexing

- ♣ Allows multiple signals to travel **simultaneously** over one medium
- ♣ **Multiplexing** is used for **most effective use** of the available **channel capacity** (bandwidth utilization) for reducing bandwidth wastage.
 - **Multiplexer (mux)** is a device that **combines** several signals into a single signal.
 - **Demultiplexer (demux)** separates the combined signals and regenerates them in original form

Types of Multiplexing



TRANSMISSION MEDIA



Transmission Media

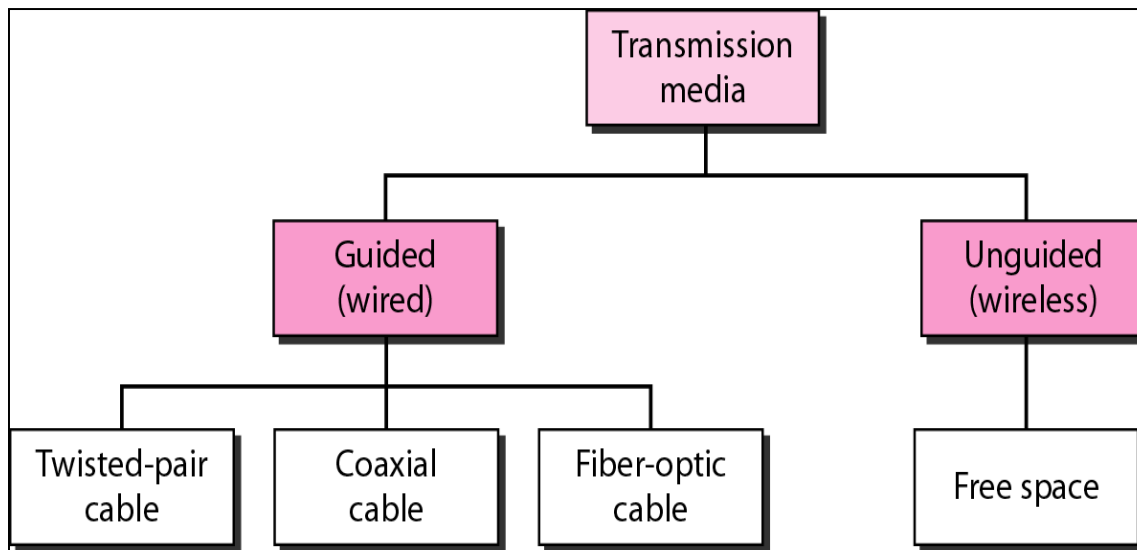
- Is **physical pathways** through which signals are transmitted from a sender to a receiver in a communication system.

General factors used in Media Selection

- *Type of network they used*
- *Cost of implementation*
- *Transmission distance they can transmit with loss of signal strength*
- *Security they have*
- *Error rates they have*
- *Transmission speed and bandwidth*
- *Tolerance of interfaces*

Types of transmission of medias/channels

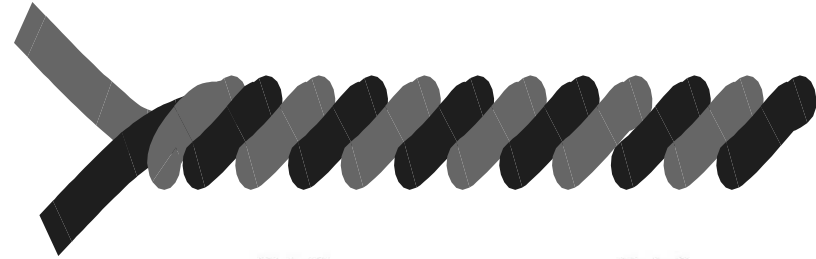
- ❖ *Guided Media*: communication follows an **exact** path
- ❖ *Unguided Media*: communication follows **no** specific path.



1. Guided Transmission Media

1. Twisted Pair (TP) Wires

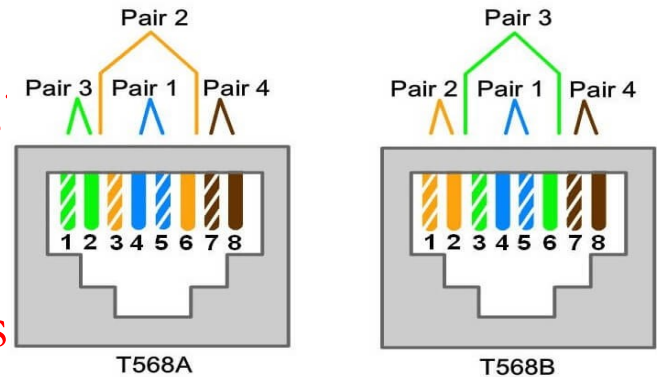
- Shielded twisted pair
- Unshielded twisted pair



❖ But what is the advantage of twisting wires

- Used for telephones and LANs

❖ TP cables have a **number** of **pairs of wires**



❖ **Telephone lines:** two pairs (4 wires used by the telephone)

❖ **LAN cables:** 4 pairs (8 wires)

❖ TP cables are **least-expensive** and most **commonly-used** transmission medium.

TYPES TP

Unshielded and Shielded TP

- ❖ Is **commonly** used for computer networks such as a building, that is, for local area networks (**LANs**).
- ❖ Connect using **RJ-45** connectors.
- ❖ Data **rates** for LANs using twisted pair today range from **10 Mbps to 100 Mbps**
- ❖ Ordinary telephone wire {**RJ-11** connector)

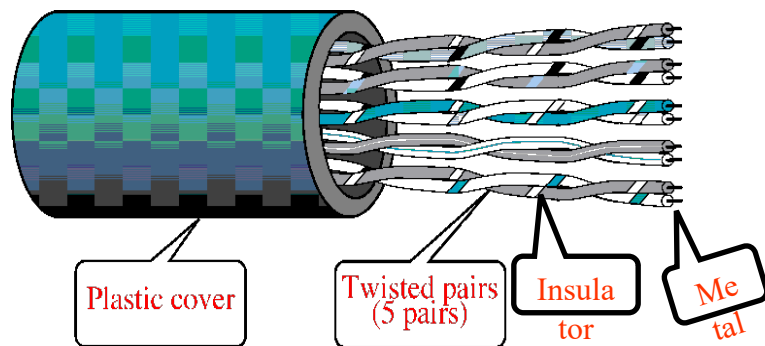


TABLE 1-2

Different UTP
Category Cabling

UTP Category	Purpose	Transfer Rate
Category 1	Voice only	
Category 2	Data	4 Mbps
Category 3	Data	10 Mbps
Category 4	Data	16 Mbps
Category 5	Data	100 Mbps
Category 5e	Data	1 Gbps (1000 Mbps)
Category 6	Data	10 Gbps

2. Coaxial cable

- ❖ Similar to cable **TV** wire
- ❖ Used for cable **television**, LANs, telephony
- ❖ **One** wire runs through cable

Types of coaxial cables

Thinnet: -refers to **RJ-58** cabling

- About $\frac{1}{4}$ inch thick
- Connected to workstation NIC using British Naval Connector (**BNC**) port connector
- Used for **short** distance communication
- Max length of thinnet is **185** meters

Thicknet : - refers **RJ-8** cabling

- About $\frac{1}{2}$ inch thick
- Used for **long** distance communication
- Max length of thicknet is **500** meters

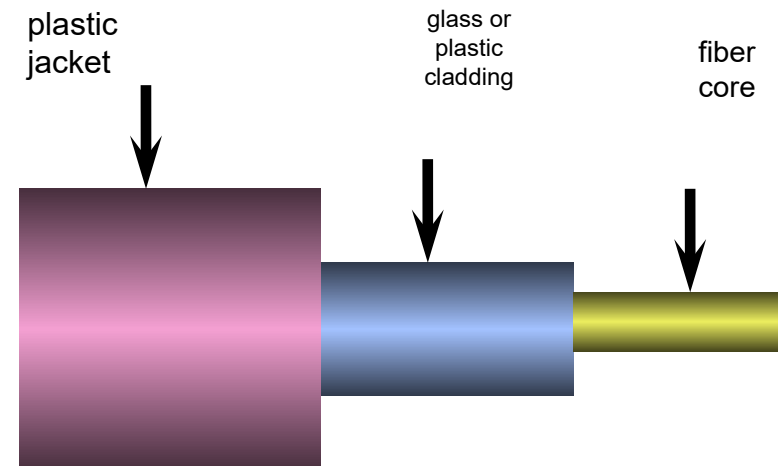
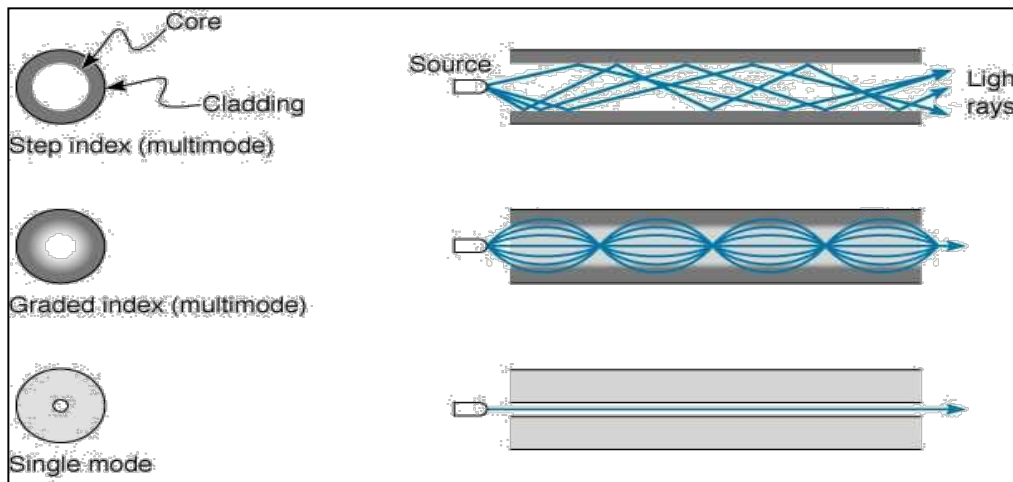


3. Fiber Optic Cable

❖ A **fiber optic cable** is a network cable that contains strands of **glass fibers** inside an insulated casing.

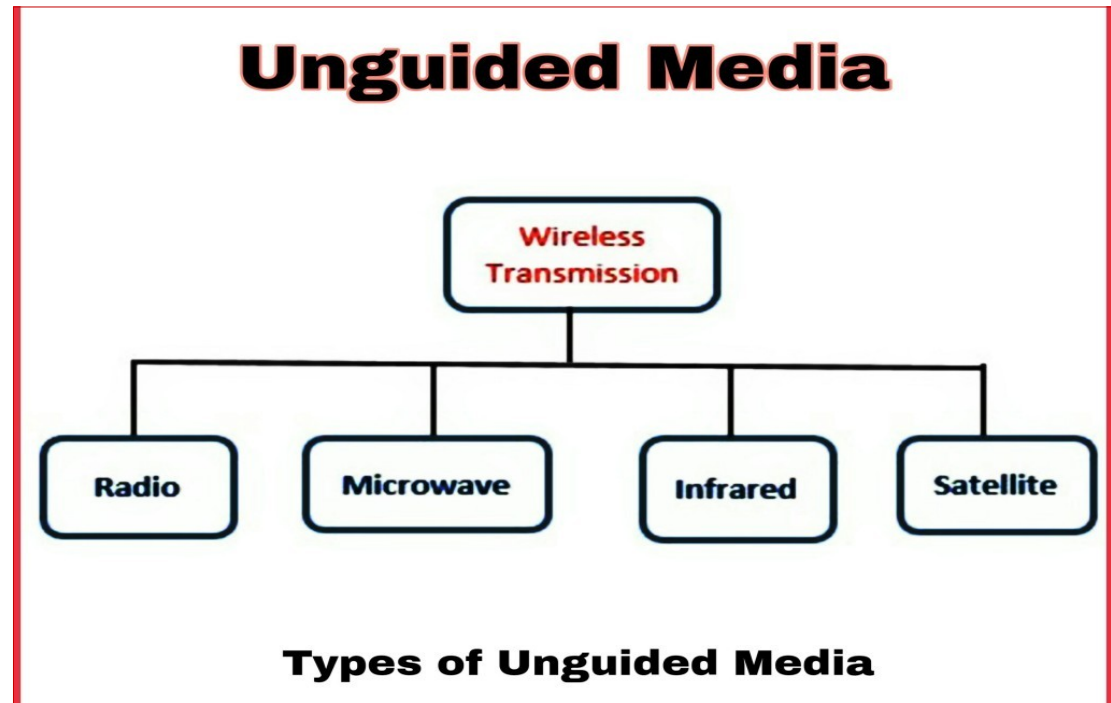
❖ **Fibers designed for**

- long-distance data transmission,
- high-performance data networking, telecommunications.
- Provides **higher bandwidth** and



2. Wireless Media/unguided Transmission Media

- **Does not use a conductor** or wire as a communication channel.
- Instead it uses the **air** or **vacuum** as medium to carry information
 - Data transmitted through the **air**
 - LANs use **radio waves**
 - WANs use **microwave signals**
 - Easy to setup
 - Difficult to secure



Cont'd

Characteristics of Radio waves transmission

- Long distance wavelength
- But Short frequency
- Frequency ranges 300Hz -300GHz (ranges 3KHz-5GHz)
- These waves are Omni directional.
- Includes AM and FM radios, cellular phones, wireless LANs (IEEE 802.11) and Bluetooth

Cont'd

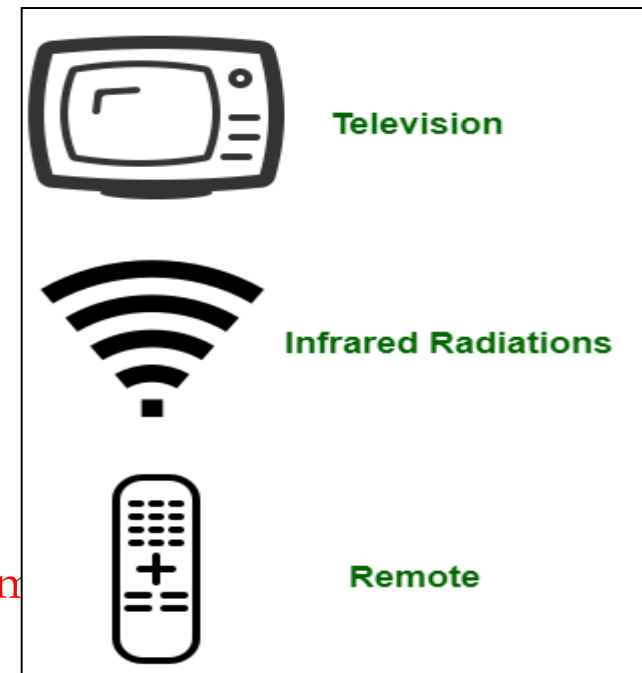
Characteristics of Microwaves waves transmission

- Microwaves have **higher** frequencies than radio waves.
- They are used for point-to-point communication in microwave links and **satellite** communication

Cont'd

❏ Characteristics of Infrared waves transmission

- ❖ Infrared Waves are the electromagnetic waves which have frequency range between 300 GHz to 400 GHz.
- ❖ These cannot travel along long distances.
- ❖ Used for short range communication.
- ❖ They also use line-of-sight of propagation.
- ❖ Cannot pass through solid objects like walls.
- ❖ These also not penetrate through walls.
- ❖ The most common application of the IR waves is remote for TV, DVD players and stereo system.



PART 2

Computer Network Type and Topologies

OBJECTIVES:

- ❖ Recognize network Topologies and Type
- ❖ Understand network components

What is Computer Network ?

- It is a collection of interconnected devices, such as **computers**, **servers**, **printers**, **routers**, **switches**, and other hardware, that are linked together to enable communication and data sharing.
- These devices are connected by various **transmission media**, such as **wired** or **wireless** connections.

Types of computer network

Computer networks can be categorized based on:

- I. Based on **geographical coverage** as:

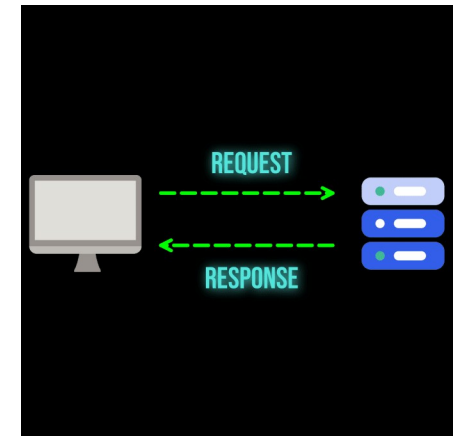
1. **Local Area Network (LAN):** A LAN covers a **small** geographical area, such as a **home, office building, or campus**. It typically consists of interconnected devices within a limited space.
2. **Metropolitan Area Network (MAN):** A MAN covers a larger area than a LAN, typically spanning a **city** or metropolitan region. It connects multiple LANs together.
3. **Wide Area Network (WAN):** A WAN spans a large geographical area, often connecting multiple cities or even **countries**. The internet is a prime example of a WAN.
4. **PAN:** PAN stands for Personal Area Network.
 - It is a type of computer network that covers a very small area, typically within the immediate proximity of an individual person.
 - PANs are designed for personal use and are often centered around an individual's personal devices.

Types of computer network

Computer networks can be categorized based on:

- II. Based on **Network Architecture, security and access** as:

1. **Client-Server Network:** In a client-server network, there are dedicated servers that provide services or resources to client devices.
 - **Clients** request services, and **servers** respond to those requests.
2. **Peer-to-Peer Network (P2P):** In a peer-to-peer network, devices, or peers, share resources directly with each other without the need for a central server.
 - Each device can act as both a client and a server



Internetwork

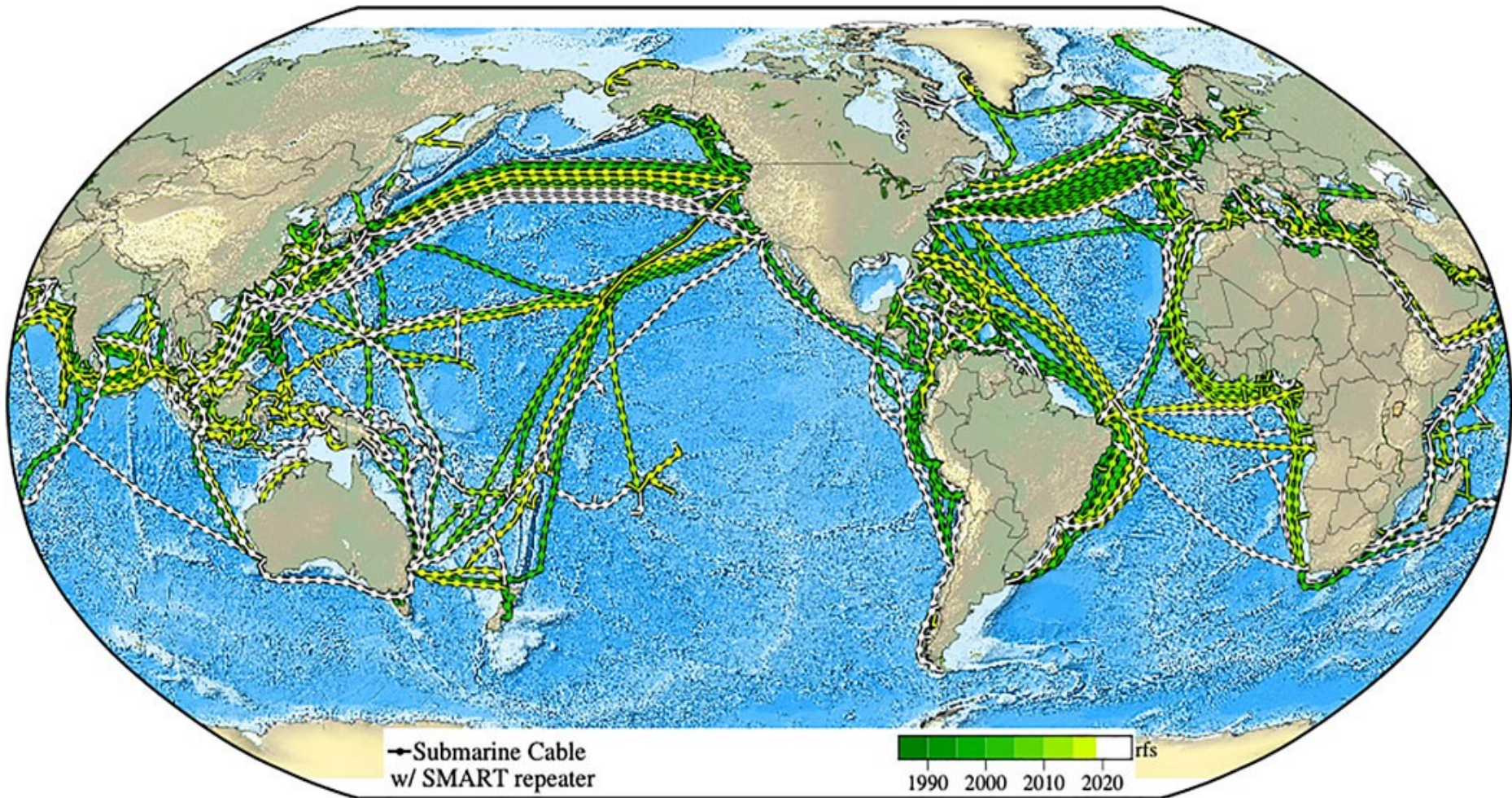
- A **network of networks** is called an internetwork, or simply the **internet**.
- An internetwork is **formed** when multiple networks are interconnected **to enable communication and data exchange** across **different** networks.
 - It allows devices on one network to communicate with devices on other networks, creating a vast global network of networks.
- Internet uses **TCP/IP protocol suite** and uses **IP** as its addressing protocol.
- Present day, Internet is widely implemented using **IPv4**.
 - Because of shortage of address spaces, it is gradually migrating from **IPv4 to IPv6**.
- It uses **WWW**

Internetwork...

- Internet uses very high-speed **backbone** of **fiber optics**.
- To inter-connect various **continents**, fibers are laid **under sea** known to us as **submarine** (subsea) communication cable.
- Internet is widely deployed on World Wide Web services using **HTML** linked pages and is accessible by client software known as **Web Browsers**



Internetwork...



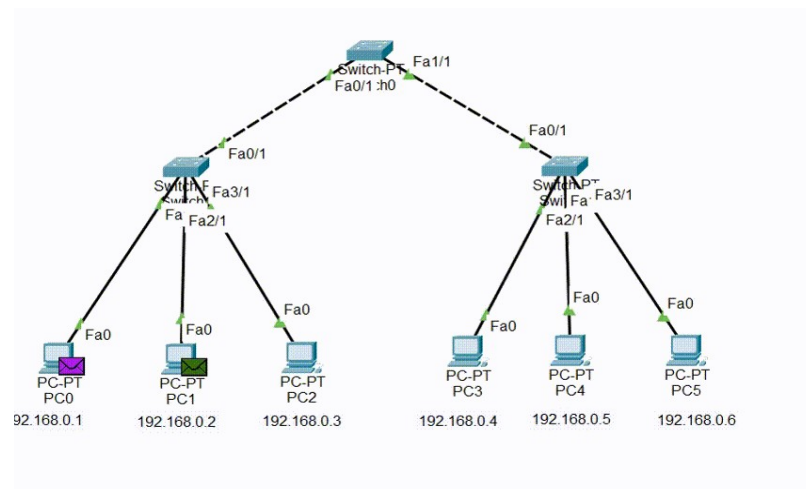
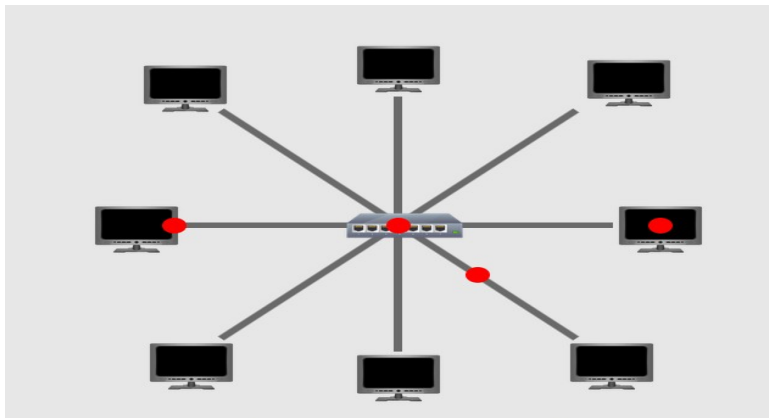
Ethiopia in which country directions they connect to submarine undersea cables

- Ethiopia, being a **landlocked** country, **does not directly** connect to submarine undersea cables as it does not have direct access to the coast.
- However, Ethiopia has established connections to international submarine undersea cables through **neighboring countries** that have coastal access.
- **Here are some examples:**
 - **Djibouti**
 - **Sudan**

WHAT IS NETWORK TOPOLOGY?

What is Network Topology?

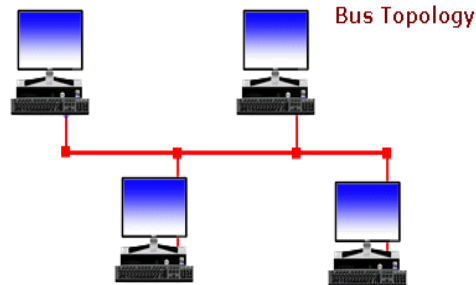
- Topology defines the **structure** of the network of how all the components are interconnected to each other.
- Refers to the manner in which the **links** and **nodes** of a network are **arranged** to relate to each other.
- it is a diagram **helps visualize the communicating devices**, which are **modeled as nodes**, and **the connections between the devices**
- There are two types of topology: **physical** and **logical** topology.



Type

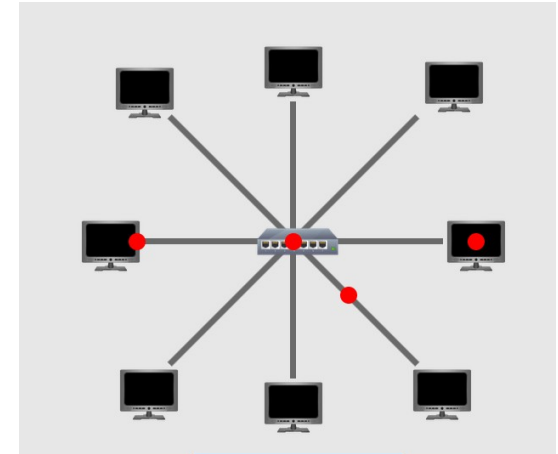
Bus topology

- is when every network node is directly connected to a main cable.



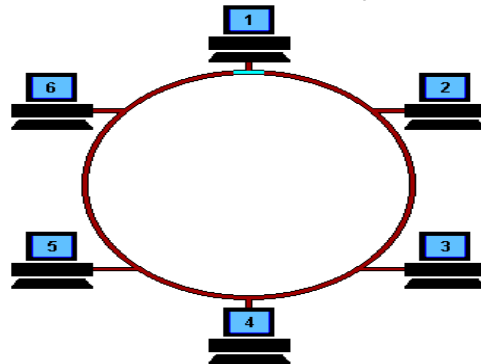
Star topology

- all nodes are connected to a single, central hub



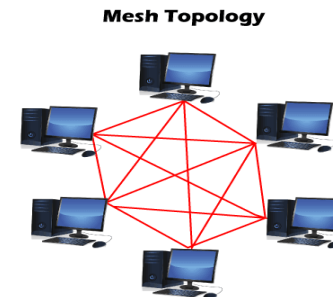
Ring topology

- nodes are connected in a loop, so each device has exactly two neighbors.



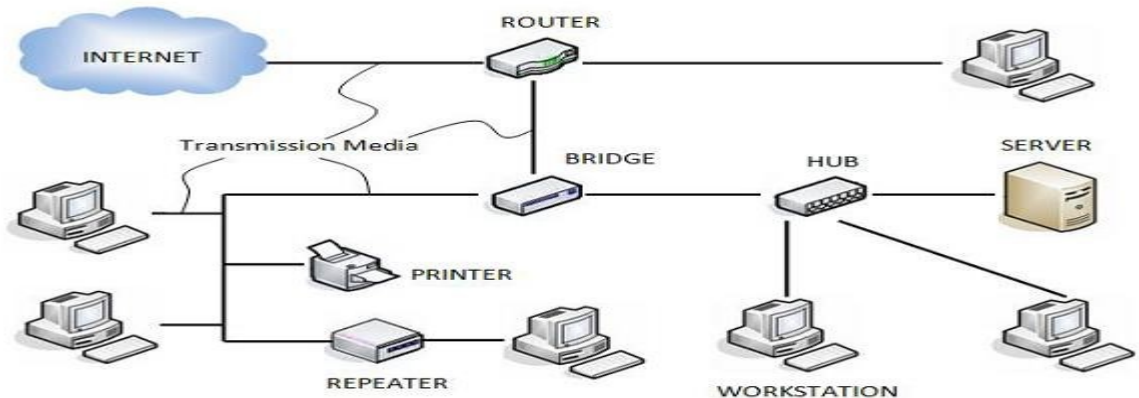
Mesh topology

- is defined by overlapping connections between nodes



Network component

- Computer networks components comprise both **physical parts** as well as the **software** required for installing computer networks, both at organizations and at home.
- The hardware components are the **server**, **client**, **peer**, **transmission medium**, and **connecting devices** (router, switch, access point, hub, gateway, repeater).
- The software components are **network operating system** and **protocols**(OSI or TCP/IP suit)

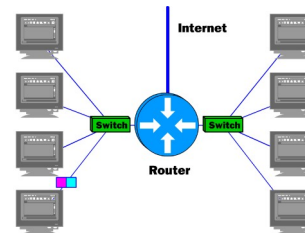
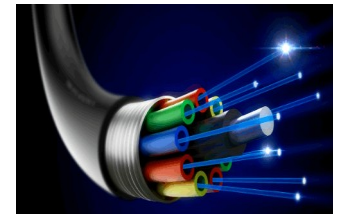


Network component...

- Here are some common network components:

1. Network Devices:

- ❖ Switches.
- ❖ Routers.
- ❖ Firewalls
- ❖ Wireless Access Points (WAPs)



2 Network Cabling:

- Ethernet Cables.
- Fiber Optic Cables.

PART 3

COMMUNICATIONS NETWORK PROTOCOLS

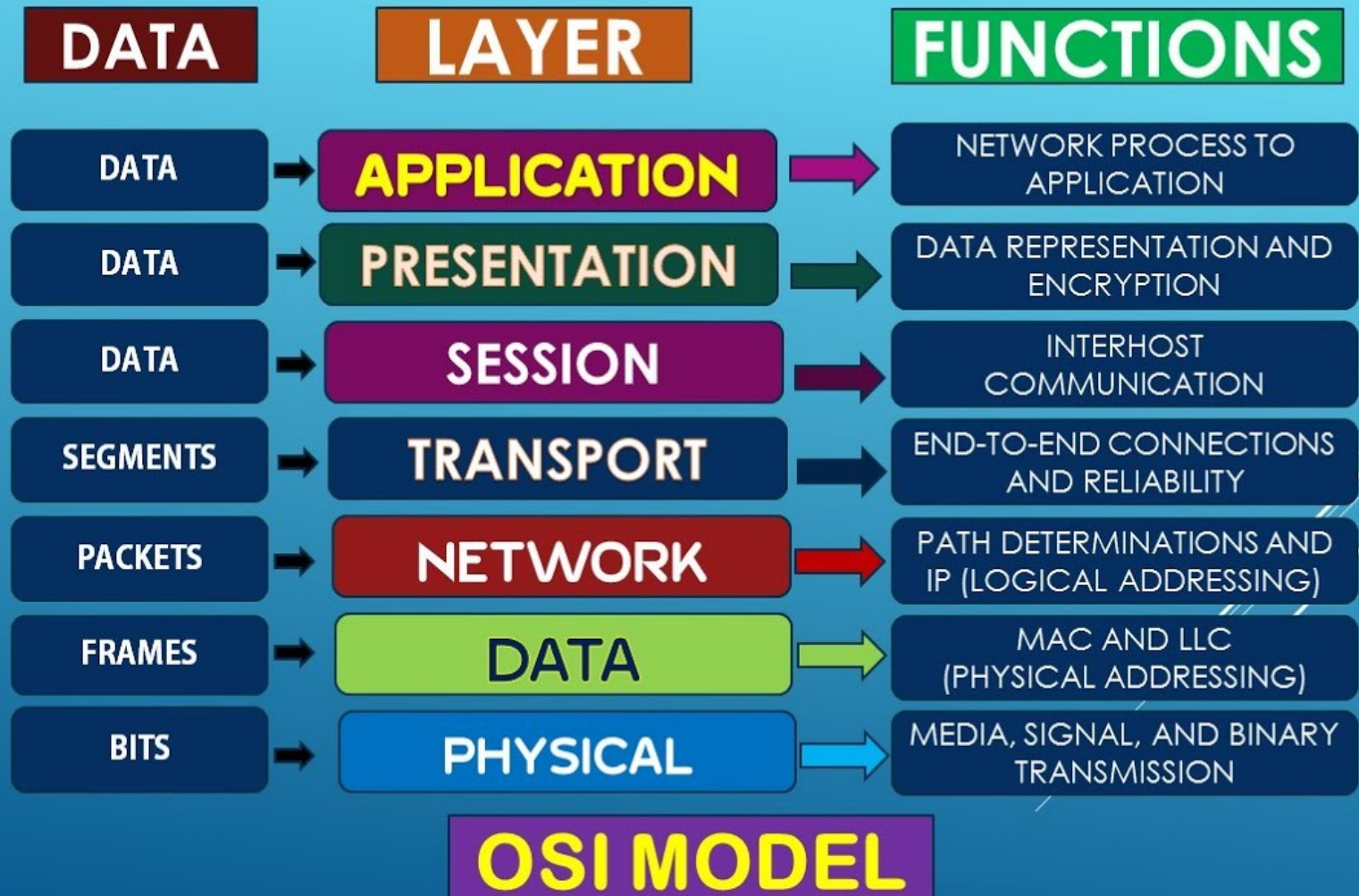
&

LAYERED MODELS

OBJECTIVES:

- ❖ Identify and understand TCP/IP and OSI Reference Model
- ❖ Understand network protocols and functions
- ❖ Design and implement IP addressing and Subnets

NETWORK LAYERING





Why TCP/IP and OSI model

- Modularity

- Easier troubleshooting

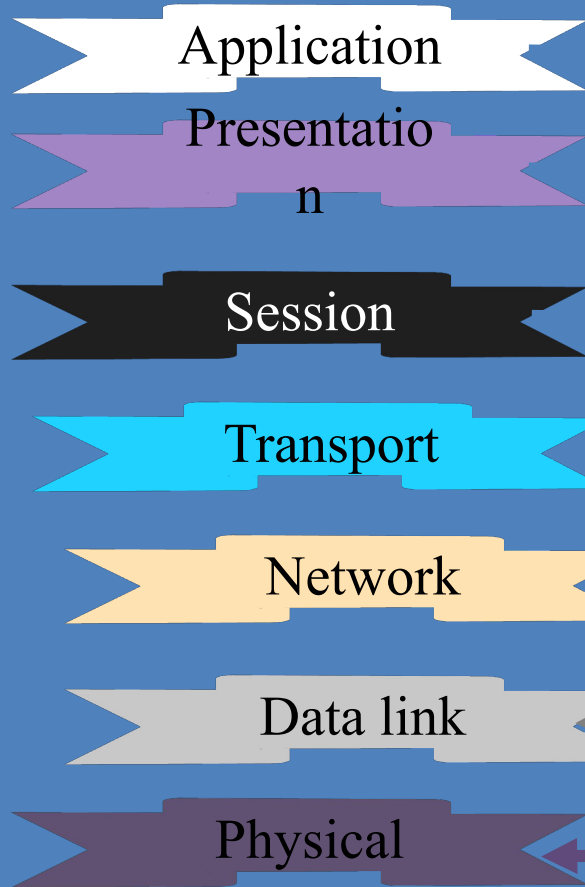
- Enhance functionality

Layering standards

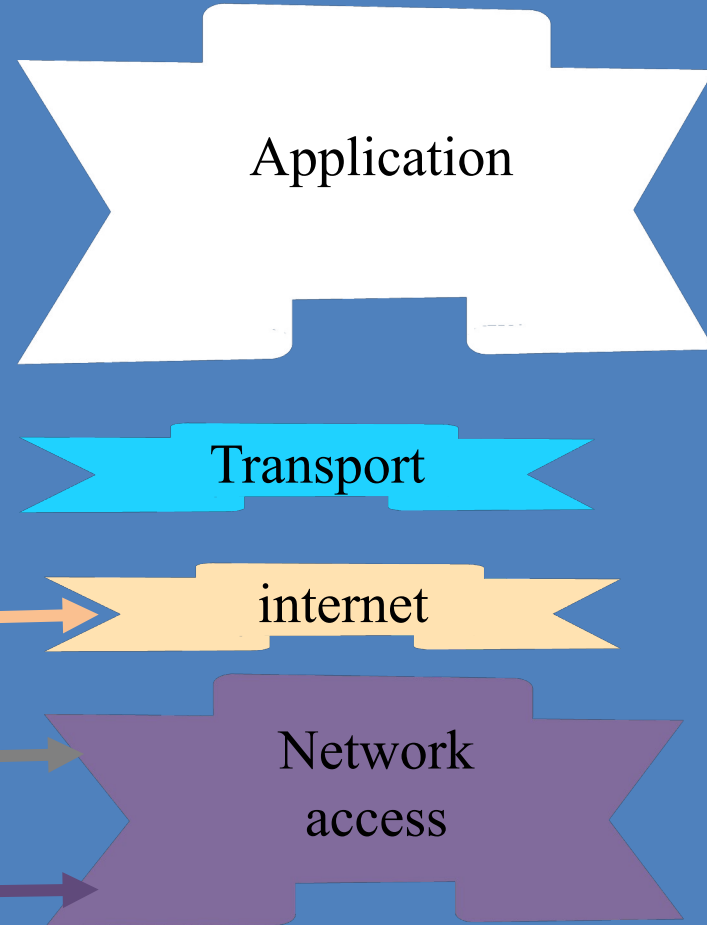
1. International Standardization Organization (OSI)
2. Layers in the TCP/IP Protocol Suite

❖ Commonly used networking models, such as the **TCP/IP model**, divide network functionality into different layers.

OSI reference model



TCP/IP

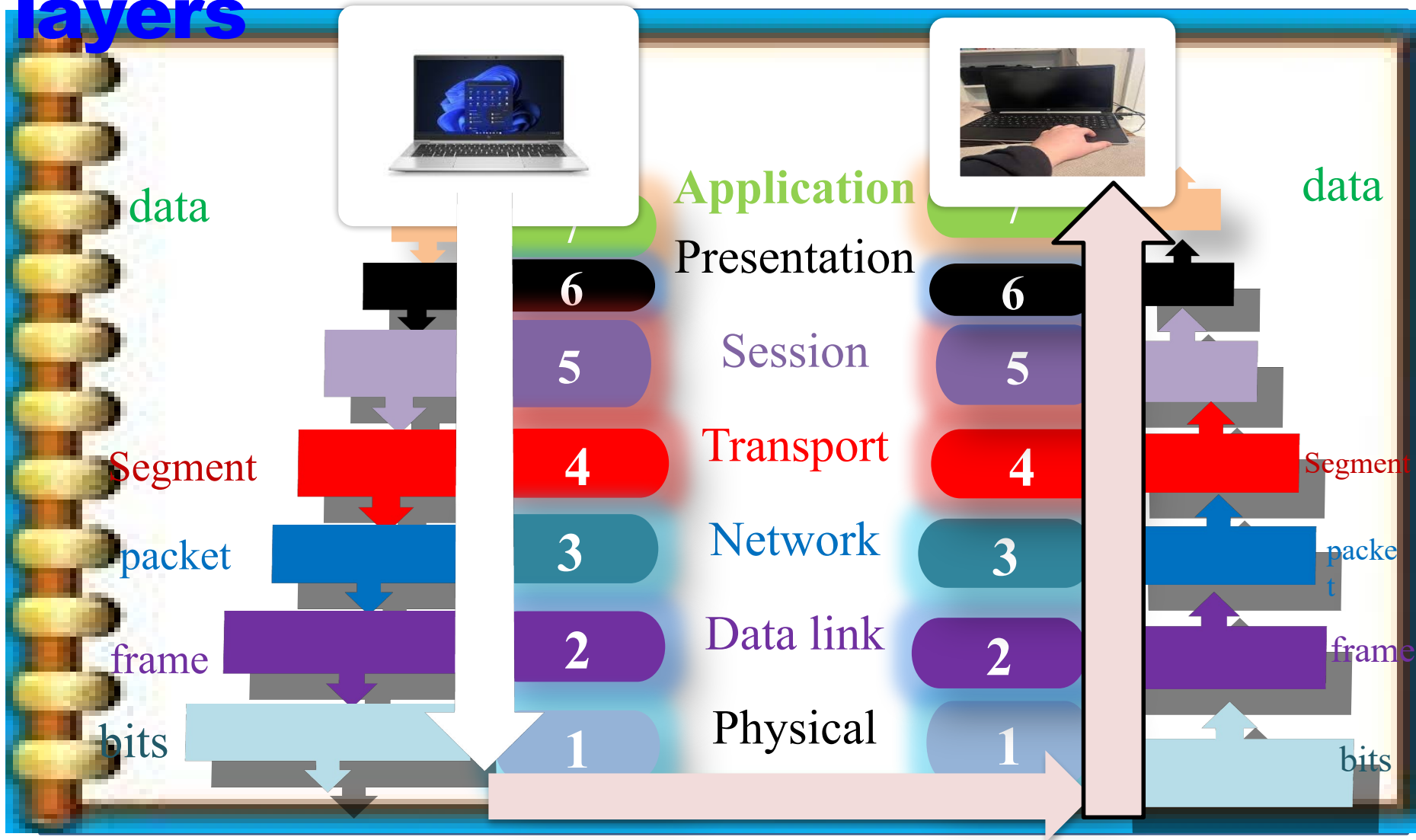


OSI Model

❖ What is OSI reference model?

- The OSI (Open Systems Interconnection) reference model is a conceptual framework that **standardizes** and **defines** how different network protocols and technologies should interact and communicate with each other.
- It consists of **seven layers**
- **However**, it is important to note that the OSI model is a reference model and **not an actual implementation**.
- Real-world network architectures, such as the Internet, often use modified or simplified versions of the OSI model, such as the **TCP/IP model**

How data travel across these 7 layers



Encapsulation

❖ Encapsulation:

- Is the process of **adding protocol-specific headers and trailers** to the **data** as it **moves down** through the layers of the OSI model.

❖ Decapsulation:

- Is the **reverse** process of encapsulation. It occurs as data **moves up** through the layers of the OSI model at the receiving end.



Encapsulation



Data=>**Segment**=>**Packet**=>frame=>bits

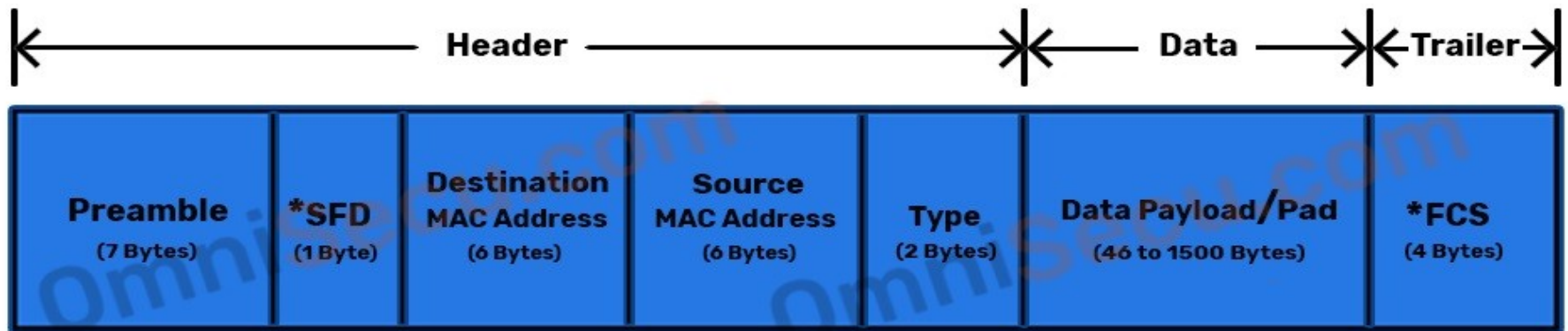


De-encapsulation



Bits->frame->packet->**segment**->**data**

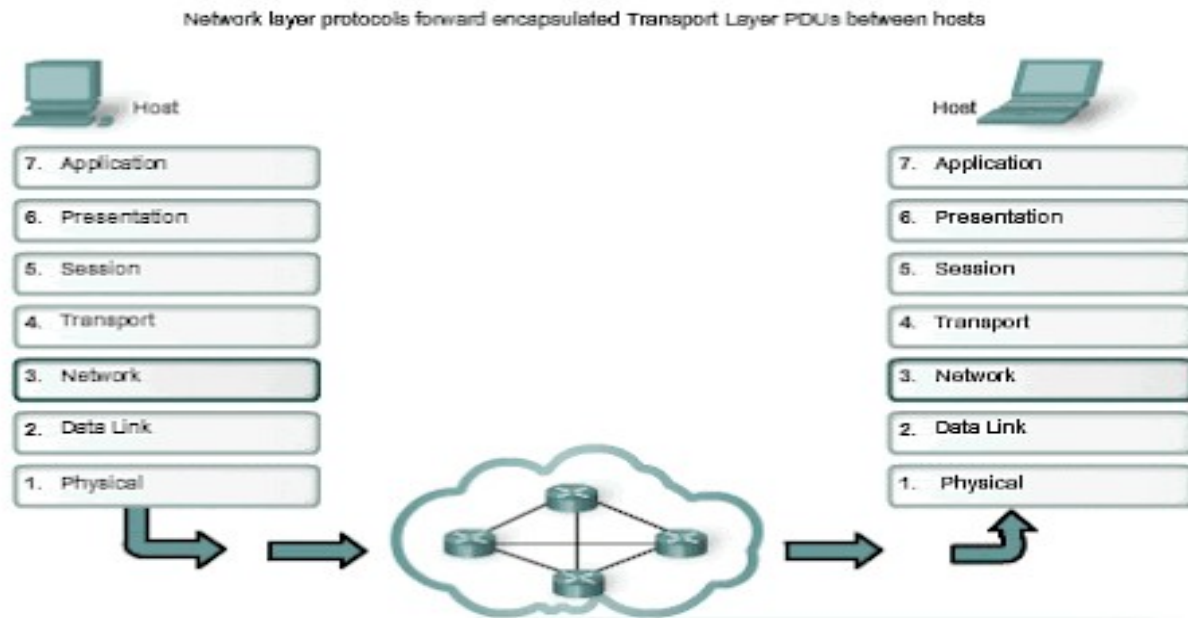
General structure of packet or frame sequence



*SFD - Start Frame Delimiter

*FCS - Frame Check Sequence

7 Layers of OSI

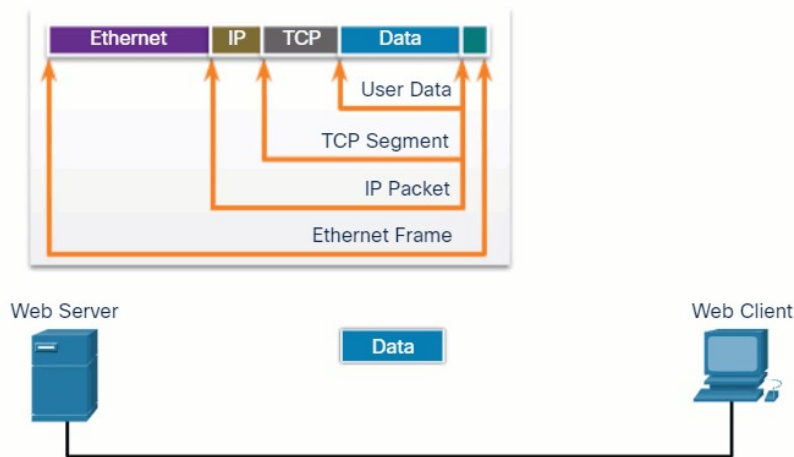


Layer 1: Physical Layer

- ❖ It define **physical characteristics of network** such as **wires, connector, voltages, data rates, Asynchronous, Synchronous** Transmission
- ❖ *Where is the position of physical layer at OSI?*
- ❖ *What did you understood from figure 4 and 5?*

^ General Responsibility of this layer are:

- **Bit signal transmission**
- **Bit synchronization**
- **Bit rate control (speed)**
- **Multiplexing**



Layer 2: Data link layer

- ❖ It received **raw bits from the physical layer** and package it into **frames**
- ❖ It provides **reliable transit of data across** a physical link by using the **MAC address**
- ❖ **Generally function of data link layers**

1. Physical Addressing:

- ❖ Uses physical addresses, such as **MAC** (Media Access Control) addresses, to identify the source and destination devices on the same network segment.

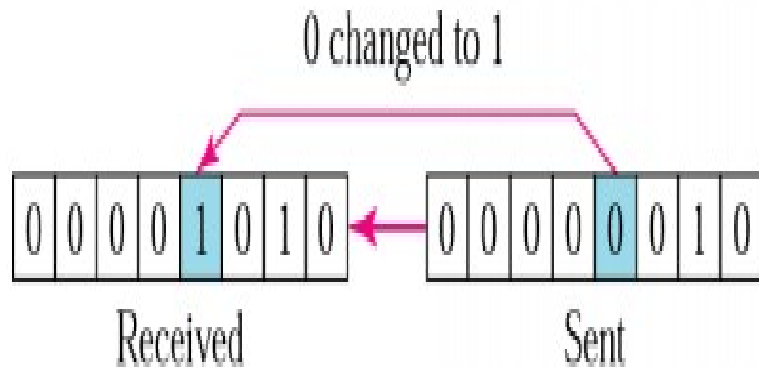
2. Framing: is process of divides the **packets** received from **network layer** in to **manageable** data units called **frame**.

3. Flow control: is mechanism controlling of **overwhelming** problem.

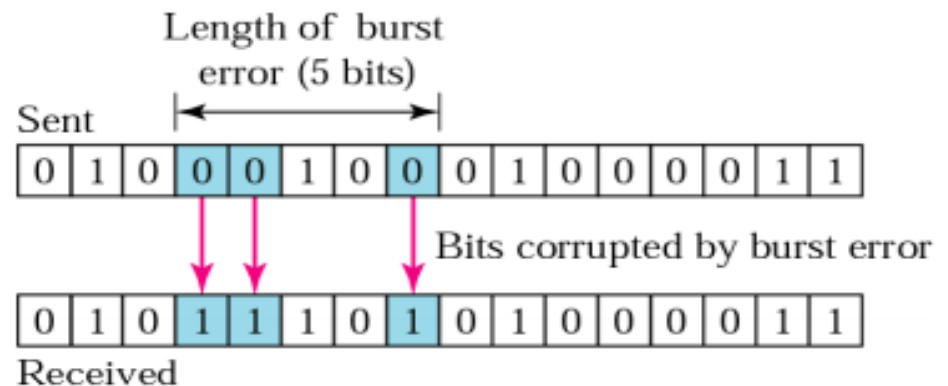
Layer 2: Data link...

4. **Error control**: is adding of reliability to the **physical layer** by adding mechanisms to **detect** and **retransmit** damaged or **lost frames**.

- It uses techniques such as **checksums** and **cyclic redundancy checks (CRC)** to verify the integrity of the received frames.



Single bit error



Burst bit error

layer 3: Network layer

1. Internetworking:

- Internetworking refers to the logical gluing of heterogeneous physical networks together to look like a single network to the upper transport and application layers.

2. Addressing

- ❖ Network layer addresses are always logical addresses which means these addresses are software based addresses which can be changed by appropriate configurations.
- ❖ These addresses are called IP address which can be IPv4 or IPv6

3. Packetizing:

- ❖ is encapsulation of packets received from upper-layer protocols and makes new packets.

4. Routing, Routing Protocols and Multicasting

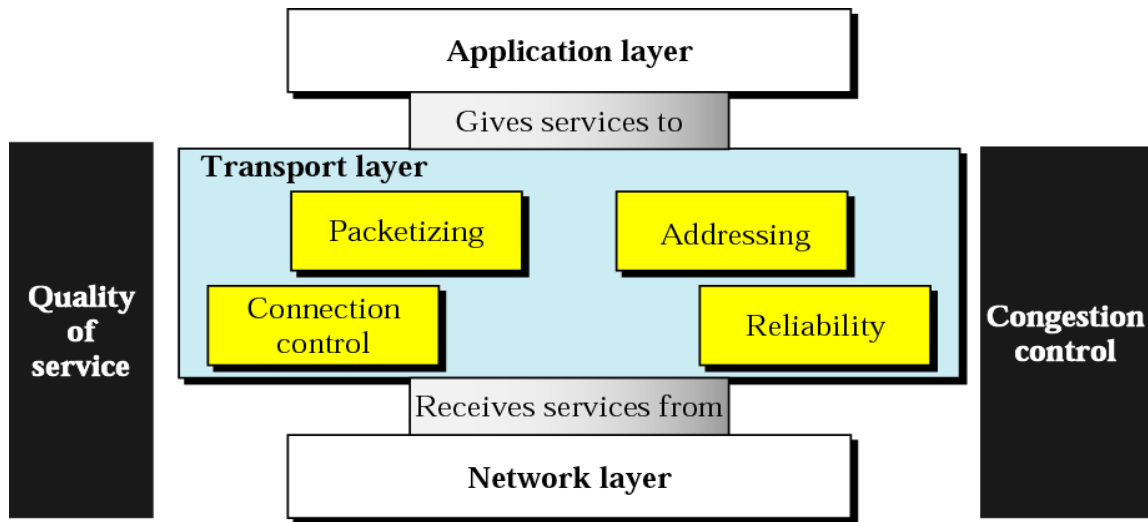
- ❖ How do you choose a route (path) when you drive?

5. Fragmentation

- ❖ Making data or frame possible to pass through other networks which is this is called fragmentation.

Layer 4: Transport Layer

- ❖ The core or the **heart** of the whole protocol hierarchy
- ❖ It provides **reliable**, **cost-effective** data transfer from the source to the destination on top of an unreliable network; process-to-process delivery



Responsibility of transport layer

1. Segmentations



Transport Layer

2. Addressing

- ❖ It is **port number** that used identify each several processes run at application layer and its length is a **16-bit unsigned integer**, and ranging from **10 – 65535**

3. Connection Control

- ❖ A transport-layer protocol can be either **connectionless** (UDP) or **connection-oriented** (TCP)

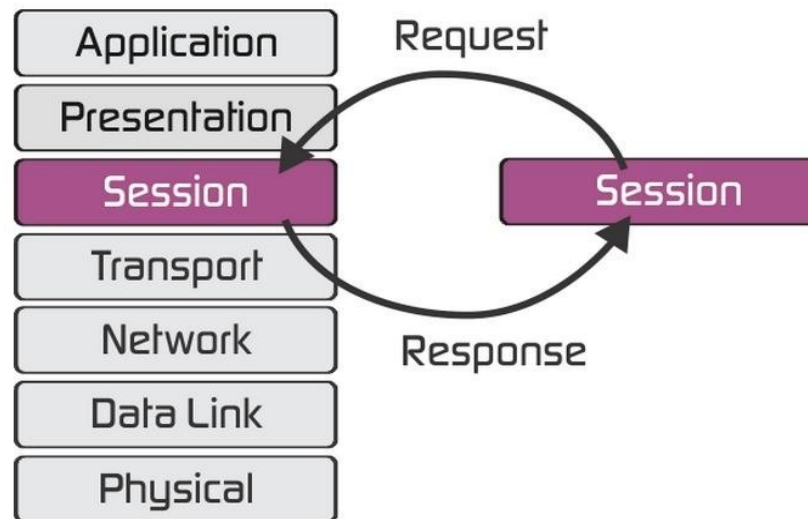
4. Reliability

- ❖ Its services **reliable** or **unreliable** b/c of UDP and TCP protocols

Port Number(s)	Protocol	Port Type
21	FTP	TCP, UDP
22	SSH	TCP, UDP
23	Telnet	TCP, UDP
25	SMTP	TCP, UDP
53	DNS	TCP, UDP
67/68	DHCP	UDP
80	HTTP	TCP, UDP
110	POP3	TCP, UDP
137-139	NetBIOS/NetBT	TCP, UDP
143	IMAP	TCP
161/162	SNMP	TCP, UDP
389	LDAP	TCP, UDP
427*	SLP	TCP, UDP
443	HTTPS	TCP, UDP
445	SMB/CIFS	TCP
548	AFP	TCP
3389	RDP	TCP, UDP
*Can also be used for AFP		

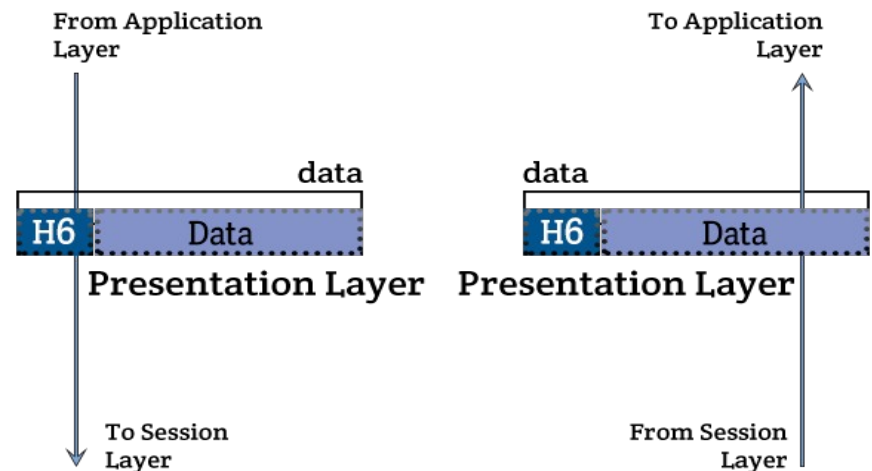
Layer 5: Session Layer

- ❖ The session layer defines how to **start, control and end conversations** (called sessions) between applications
- ❖ It is responsible for **establishing, managing and terminating** sessions between **end-user** application processes.
- ❖ It allows the systems to communicate in either **half-duplex** or **full-duplex** mode of communication.



Layer 6: Presentation Layer

- ❖ The primary responsibility is to ensure the **compatibility** of data formats
- ❖ Acts as a **translator** or **converter**, transforming data from the application layer into a format that can be understood by the receiving system.
- ❖ This layer is primarily responsible for the following tasks
 - **Data translation,**
 - **Data encryption for security purpose**
 - **Data decryption**
 - **Data compression for security purpose**
 - **Data decompression**



7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data link
1	Physical

Layer 7: Application Layer

- ❖ Application layer is responsible for providing user interfaces
- ❖ Interacts directly with the end-user and provides a means for users to access network services and resource

1. Service providing

- ❖ It support services such as email, remote file access and transfer.
- ❖ It facilitates the user to use the services of the network.
- ❖ It is used to develop network-based applications such as telnet, DHCP, FTP, SNMP.

Application Layer (Layer 7)...

2. The Client-Server Model

- ❖ Computer **runs a program** to either **request** a service from another computer (**client**) or to provide a service to another computer (**server**)
- ❖ **Examples** of servers include **web servers**, **mail servers**, and **file servers**.

3. Addressing

- ❖ A **client** and a **server** communicate with each other **using addresses**
- ❖ Some examples of application layer address or they are server name
 - **Mail** server address: <http://mail.amu.edu.et>
 - **SMIS** server address: <http://smis.amu.edu.et>
 - **Web** page (host name): <http://www.amu.edu.et>
- ❖ So, application layer protocols have responsible to map these address with its respective IP address with DNS.
- ❖ **DNS** is server or service provided at application layer to resolve IP addresses from **hostnames (URL)**,

Summary of OSI layer's responsibilities

1. Physical layer

- Bit signal transmission
- Bit synchronization
- Bit rate control (speed)
- Multiplexing

4. Transport layer

- Segmentation
- Addressing
- Flow control
- Error detection and correction

2. Data link layer

- Framing
- Physical addressing
- Flow control
- Error detection and correction

5. Session layer

- Connection establish
- Connection management
- Connection termination

7. Application layer

- Service providing
- Addressing
- Client-server modeling

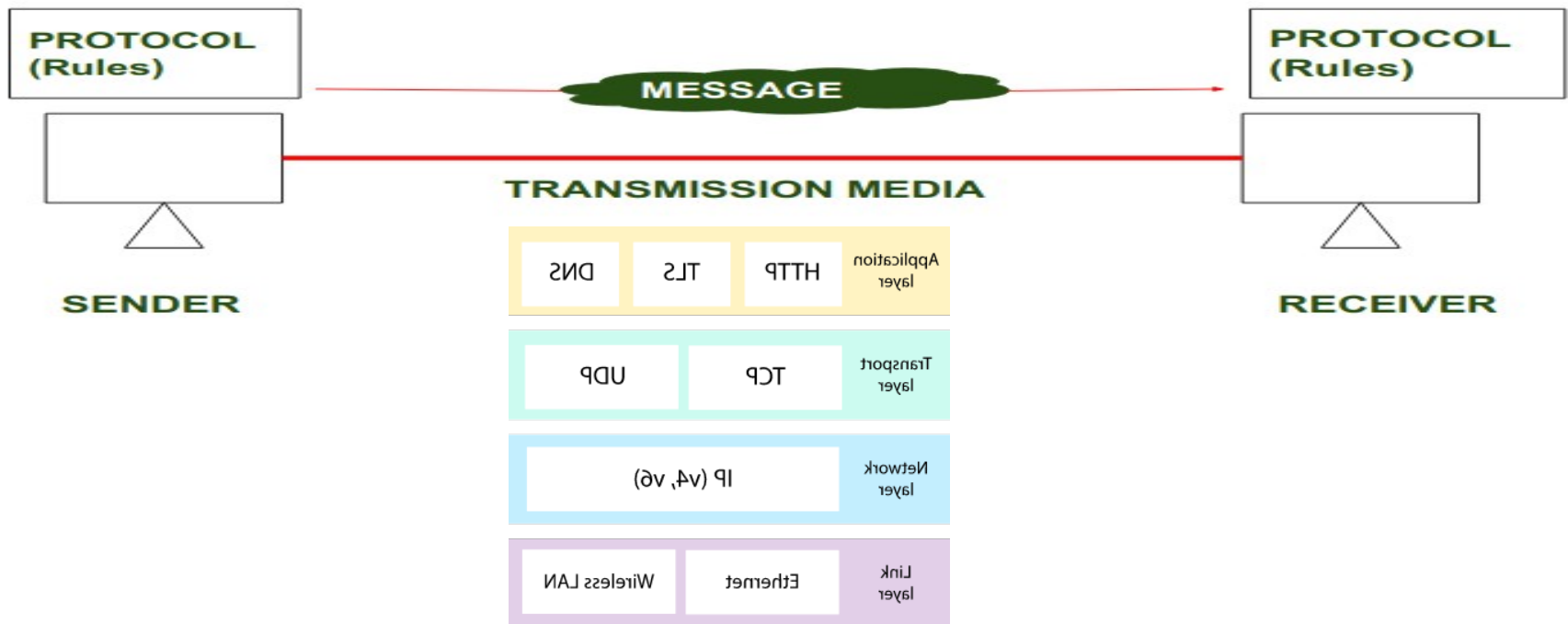
3. Network layer

- Internetworking
- Packetization
- Fragmentation
- Defragmentation
- Addressing
- Sequencing
- Error notification

6. Presentation layer

- Security encryption
- Translation
- Compression
- Decompression

Network Protocols Standards and Functions





What is Protocols?

- ♣ A **protocol** is basically a synonym for the **rule**.
- ♣ In Computer Networks, basically, **communication** occurs between entities in different systems.
- ♣ An **entity** is anything that is capable of **sending or receiving** information.
- ♣ Any two entities cannot simply send bitstreams to each other and **expect to be understood**.

Cont...

- ❖ The entities must need to agree on a protocol in order of occurrence of the communication.
- ❖ A **Protocol** is a set of rules that mainly govern data communications.
- ❖ The protocol mainly defines what is communicated, how it is communicated, and when it is communicated.

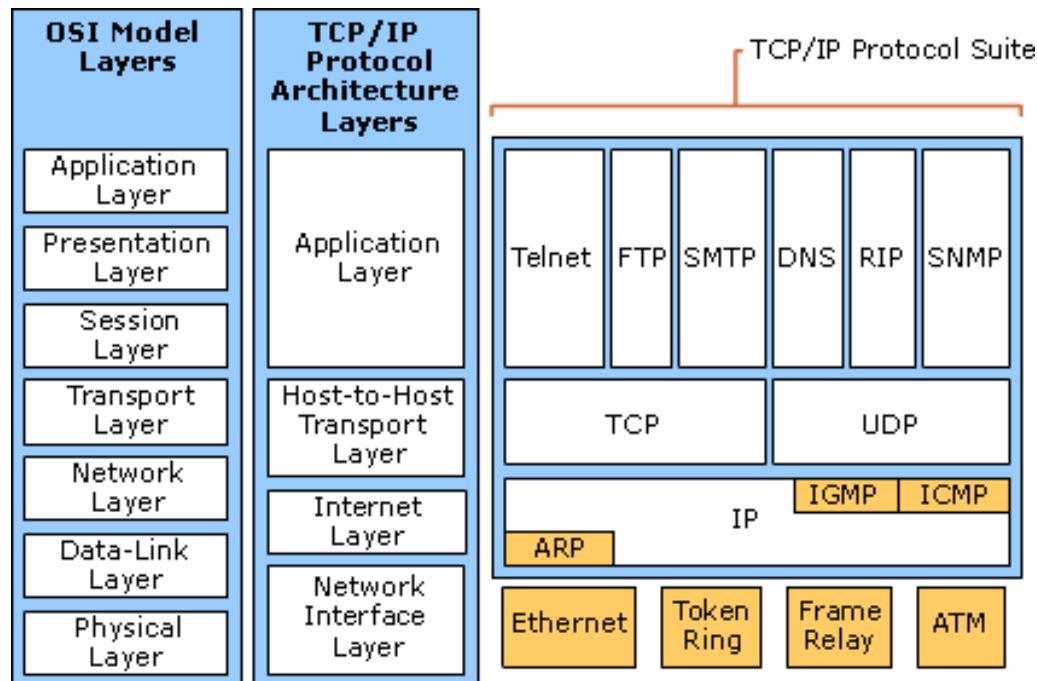
Standards

- ❖ Standards are essential in creating and maintaining an open and creative market for the equipment manufacturers and they also guarantee the national and international interoperability of the data, telecommunication technology, and process.

e.g.

- ***Institute of Electrical and Electronics Engineers(IEEE)***
- ***International Organization of Standardization(ISO)***

PROTOCOL IN EACH LAYERS



Network layer protocols

Internet Control Message Protocol -ICMP

- ❖ Used to report errors with delivery of IP data.

- ❖ Examples: of error reports by ICMP

IGMP- Internet Group Management Protocol

- ❖ Is a protocol that provides a way for an Internet computer to report its multicast group membership to adjacent routers.

ARP(Address resolution protocol)

- Sender knows logical address of receiver but not its physical address
- ARP maps 32-bit logical (IP) address to 48-bit physical address.

RARP (Reverse Address Resolution Protocol)

- RARP is a TCP/IP protocol that allows any host to obtain its IP address from the server.

Transport layer protocol

TCP(Transmission Control Protocol)

- ❖ Numbering and sequencing each segment
- ❖ TCP waits for an **acknowledgment** of the receiving end's.
- ❖ Retransmitting any segments that **aren't acknowledged**.
- ❖ It is **reliable protocols** which means provide guaranteed delivery
- ❖ It is **Connection oriented** means that a virtual connection is established before any user data is transferred.



User Datagram Protocol (UDP)...

- It **doesn't** follow segments.
- Because of this, it's referred to as an **unreliable** protocol.
- but it is **effective** protocol even if it is unreliable
- **Connectionless protocol**.
 - **Lack** of flow control and error control
 - **Faster** and efficient
- **Never pit in right order** when segment arrive at destination.

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data link
1	Physical

Application layer Protocols

Simple Mail Transfer Protocol (SMTP)

- Governs the **transmission** of **mail messages and attachments**
- **SMTP** is used in the case of **outgoing** messages
- More powerful protocols such as **POP3** and **IMAP4** are needed and available to manage **incoming messages**

Telnet:

- It allows a user on a **remote** client machine

File Transfer Protocol (FTP)

- ❖ FTP's functions are limited to listing and manipulating **directories, typing file contents, and copying files between hosts.**

Simple Network Management Protocol (SNMP)

- This protocol can also stand as a **watchdog** over the network, quickly notifying managers of any sudden turn of events.

Hypertext Transfer Protocol (HTTP)

- It's used to manage communications between **web browsers and web servers** and opens the right resource when you click a link, wherever that resource may actually reside.



- Your browser can understand what **you need when you enter a Uniform Resource Locator (URL)**, which we usually refer to as a **web address**, e.g. <http://www.google.com/>

Hypertext Transfer Protocol Secure (HTTPS)



- Hypertext Transfer Protocol Secure (HTTPS) is also known as **Secure Hypertext Transfer Protocol**.
- It uses **Secure Sockets Layer (SSL)**.

Dynamic Host Configuration Protocol (DHCP)

- IP address can be assigned **manually** or **dynamically**.
- Manual is done by **administrators**, where as **server** assign IP dynamically.
 - DHCP** assigns IP addresses to hosts dynamically.

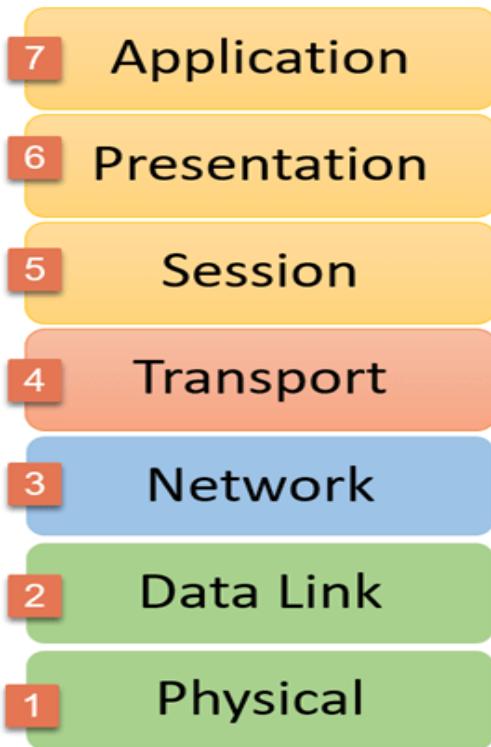
Read the following From Manuals

□ Protocols and their function for the following OSI layers

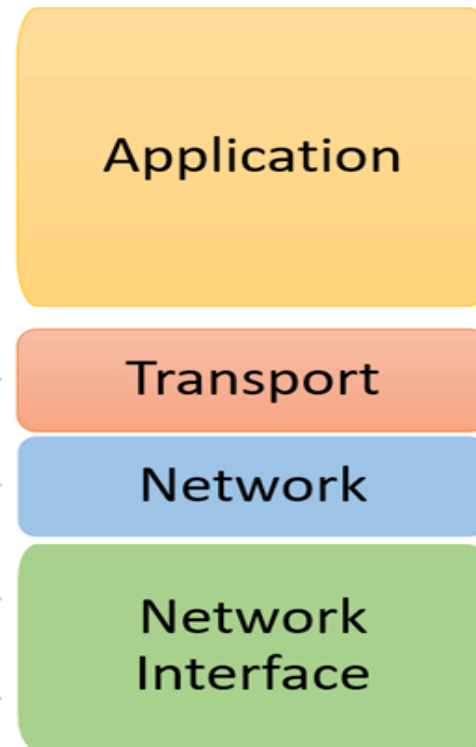
1. Physical layer
2. Data link layer
3. Session layer
4. Presentation layer

Layers in the TCP/IP Protocol Suite

OSI Reference Model

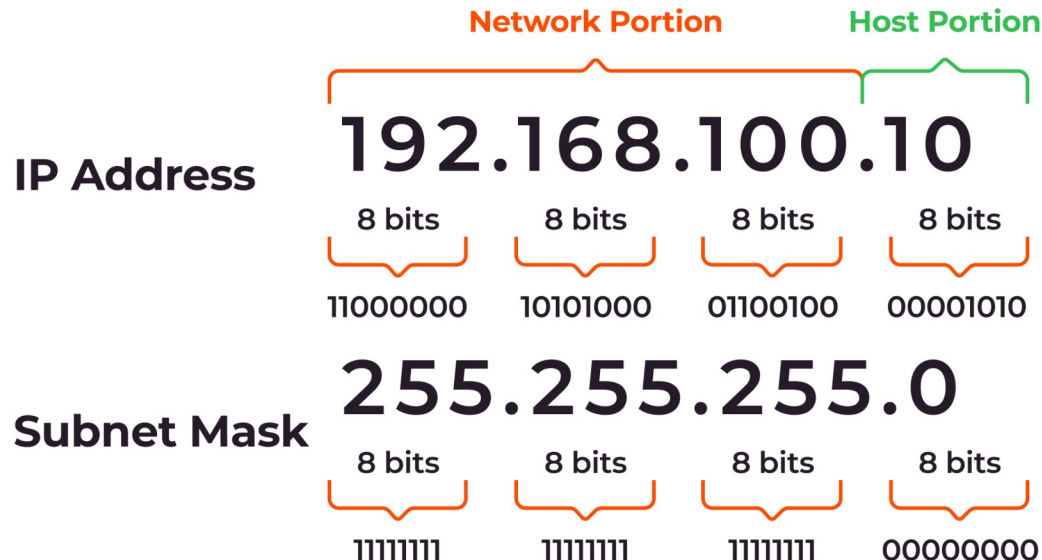


TCP/IP Conceptual Layers



IP Addressing and

Binary Notation of IP Address and Subnet



Classes of IP address

	8 bits	8 bits	8 bits	8 bits
Class A:	Network	Host	Host	Host
Class B:	Network	Network	Host	Host
Class C:	Network	Network	Network	Host
Class D:	Multicast			
Class E:	Research			

Classes	Range	N-Id & H-Id
Class A	0 to 126	1 N-Id & 3 H-Id
Class B	128 to 191	2 N-Id & 2 H-Id
Class C	191 to 223	3 N-Id & 1 H-Id
Class D	224 to 239	Use for Multicasting
Class E	240 to 255	Use for Reserch
NOTE: 127 IP address is a local Server .		

IP Classes (continued)

Network Address Range: **Class A**

- The designers of the IP address scheme decided that the **first bit of the first byte in a Class A network address must always be off, or 0.** **0**xxxxxxx
- This means a Class A address must be between **0 and 127** in the first byte, inclusive.
- So, a Class A network is defined in the first octet between 0 and 127, and it **can't be less or more.**
- Understand that **0 and 127 are not valid** in a Class A network because they're **reserved addresses.**
- The Class A format is as follows:

Network.Node.Node.Node

Class B

- In a Class B network, the RFCs (Request for Comments) state that **the first bit of the first byte must always be turned on** but the **second bit must always be turned off**.

10xxxxxx

- A Class B network is defined when the first byte is configured **from 128 to 191**.
- The first **2 octet** are assigned to the network address and the remaining **2 octet** are used for node addresses.

The format is as follows:

Network.Network.Node.Node

IP Classes

(continued)

Binary Place Values

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

Decimal
Equivalent

Description

0	0	0	0	0	0	0	0	= 0	Subnet identifier
0	0	0	0	0	0	0	1	= 1	Bottom of Class A range
0	1	1	1	1	1	1	0	= 126	Top of Class A range
0	1	1	1	1	1	1	1	= 127	Loopback address

Class A addresses begin with a number between 1 and 126

Binary Place Values

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

Decimal
Equivalent

Description

1	0	0	0	0	0	0	0	= 128	First Class B address
1	0	1	1	1	1	1	1	= 191	Last Class B address

Class B addresses begin with a number between 128 and 191

Class C

- For Class C networks, the RFCs define the first 2 bits of the first octet as always turned on, but the third bit can never be on.

110xxxxx

So, if you see an IP address that starts at 192 and goes to 223, you'll know it is a Class C IP address.

- The first 3 octet of a Class C network address are dedicated to the network portion of the address, with only 1 byte remaining for the node address.

Here's the format:

Network.Network.Network.Node

Con

t

Class C

- Addresses are assigned to groups that do not meet the qualifications to obtain Class A or B addresses
- Each Class C address supports 254 hosts
- Which is reserved for numerous networks with small number of nodes.

Binary Place Values								Decimal Equivalent	Description
128	64	32	16	8	4	2	1		
1	1	0	0	0	0	0	0	= 192	First Class C address
1	1	0	1	1	1	1	1	= 223	Last Class C address

Class C addresses begin with numbers between 192 and 223

Private IP Addresses

- These addresses can be used on a private network, but **they're not routable through the Internet.**
- Is a range of non-internet facing IP addresses used in an internal network
- This is designed for the purpose of creating a measure of **well-needed security**, but it also conveniently **saves valuable IP address space.**
- Private IP addresses are provided by network devices, such as **routers**, using network address translation.
- **Network Address Translation (NAT)**, which basically takes a private IP address and converts it for use on the Internet.

Public IP Address

- A Public IP address is the one which is globally recognizable and our ISP (Internet Service Provider) or IANA provides the same.
- Public IP identifies our home network to the outside world.
- It is an IP address that is unique throughout the entire Internet.
- Each customer who uses the Global/Public IP Block needs to pay the Service Provider or IANA.

Private IP	Public IP
Used with LAN or Network	Used on Public Network
Not recognized over Internet	Recognized over Internet
Assigned by LAN administrator	Assigned by Service provider / IANA
Unique only in LAN	Unique Globally
Free of charge	Cost associated with using Public IP
Range – Class A -10.0.0.0 to 10.255.255.255 Class B – 172.16.0.0 to 172.31.255.255 Class C – 192.168.0.0 – 192.168.255.255	Range – Class A -1.0.0.0 to 9.255.255.255 11.0.0.0 – 126.255.255.255 Class B -128.0.0.0 to 172.15.255.255 172.32.0.0 to 191.255.255.255 Class C -192.0.0.0 – 192.167.255.255 192.169.0.0 to 223.255.255.255

Network Addressing

- IP addresses identify both the network and the host
 - The division between the two is not specific to a certain number of octets
- Subnet mask
 - A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s.
 - Indicates how much of the IP address represents the network or subnet
 - The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address
- Standard (default) subnet masks:
 - Class A subnet mask is 255.0.0.0
 - Class B subnet mask is 255.255.0.0
 - Class C subnet mask is 255.255.255.0

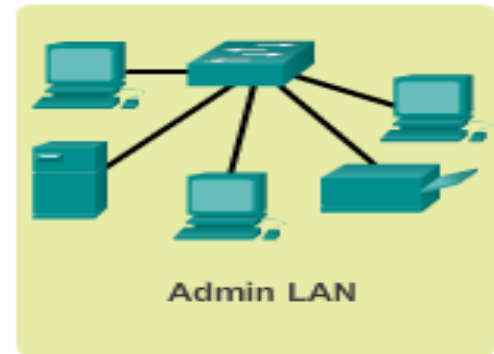
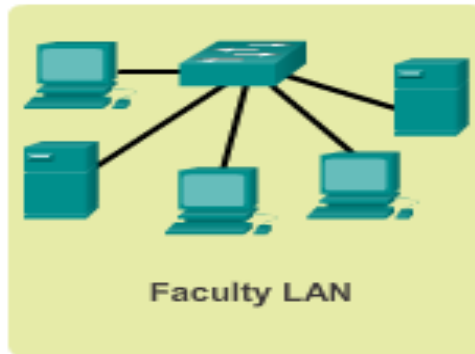
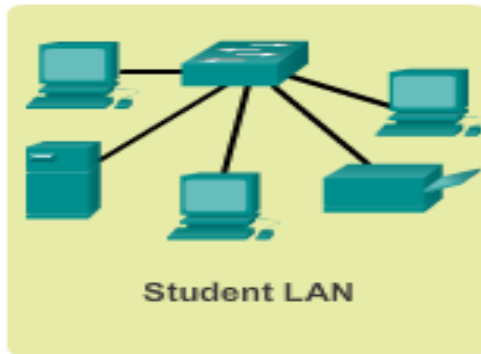
Subdividing IP Classes

- **Subnetting** - process of segmenting a network into multiple smaller network spaces called sub-networks or **Subnets**.
- The act of **taking bits from the host portion** of the address and reserving them to define the subnet address instead .

Steps to create subnet:

1. Determine the number of required **host IDs per subnet**
2. Determine the number of required **network IDs**

IP SUBNETTING IS FUNDAMENTAL



Planning requires decisions on each subnet in terms of size, the number of hosts per subnet, and how host addresses will be assigned.

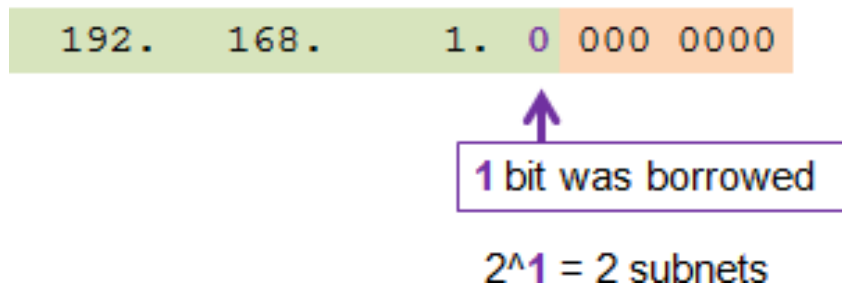
BASIC SUBNETTING

- **When network administrators create subnets**
 - They **borrow bits from the original** host field to make a set of subnetworks
 - The **number of borrowed bits determines** how many subnetworks and hosts will be available

SUBNETTING FORMULAS

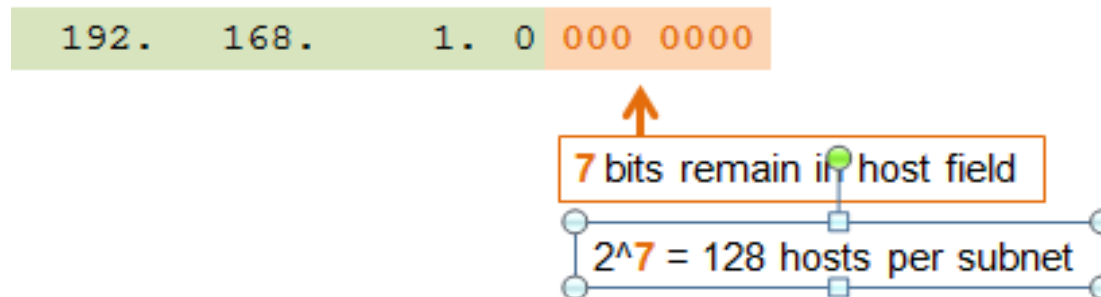
- Calculate Number of Subnets

Subnets = 2^n
(where n = bits borrowed)



- Calculate Number of Hosts

Hosts = 2^n
(where n = host bits remaining)



SUBNETTING BASED ON HOST REQUIREMENTS

There are **two** considerations when planning subnets:

1. Number of Subnets required
2. Number of Host addresses required

Formula to determine number of useable hosts

$$2^n - 2$$

- 2^n (where n is the number of host bits remaining) is used to calculate the number of hosts -2 .
- Subnetwork ID and broadcast address cannot be used on each subnet

Calculate number of subnets

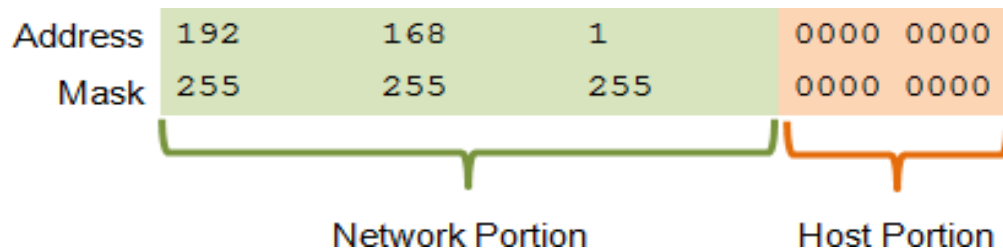
- Formula

$$2^n$$

(where n is the number of bits borrowed)

BASIC SUBNETTING

- Borrowing Bits to Create Subnets
- Borrowing 1 bit $2^1 = 2$ subnets



Original	192.	168.	1.	0	000	0000	Network 192.168.1.0/24
Mask	255.	255.	255.	0	000	0000	Mask: 255.255.255.0

Borrowing 1 Bit from the host portion creates 2 subnets with the same subnet mask

Subnet 0 ?

Subnet 1 ?

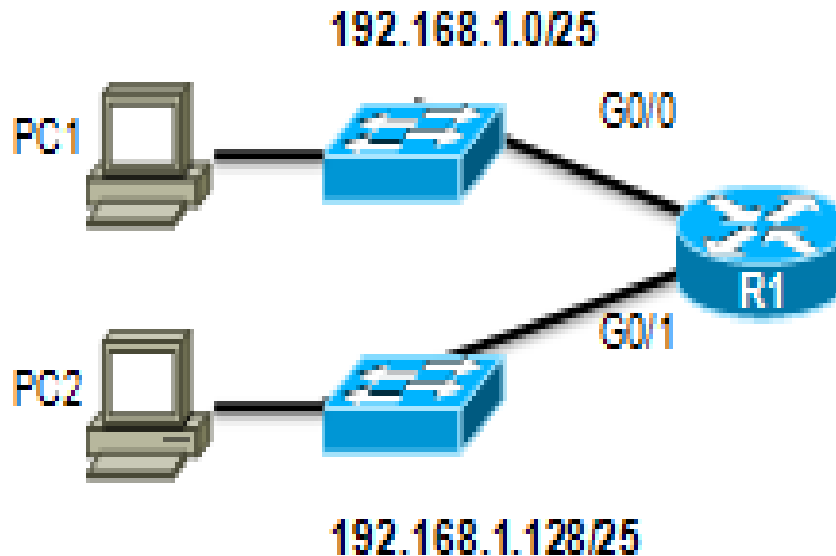
Subnet Mask ?

SUBNETS IN USE

$32-25 = 7$, $2^7 = 128$ host on each subnet

Subnet 0

Network 192.168.1.0-127/25



Address Range for 192.168.1.0/25 Subnet

Network Address

192. 168. 1. 0 000 0000 = 192.168.1.0

First Host Address

192. 168. 1. 0 000 0001 = 192.168.1.1

Last Host Address

192. 168. 1. 0 111 1110 = 192.168.1.126

Broadcast Address

192. 168. 1. 0 111 1111 = 192.168.1.127

Address Range for 192.168.1.128/25 Subnet

Network Address

192. 168. 1. 1 000 0000 = 192.168.1.128

First Host Address

192. 168. 1. 1 000 0001 = 192.168.1.129

Last Host Address

192. 168. 1. 1 111 1110 = 192.168.1.254

Broadcast Address

192. 168. 1. 1 111 1111 = 192.168.1.255

CREATING 4 SUBNETS

- Borrowing 2 bits to create 4 subnets. $2^2 = 4$ subnets

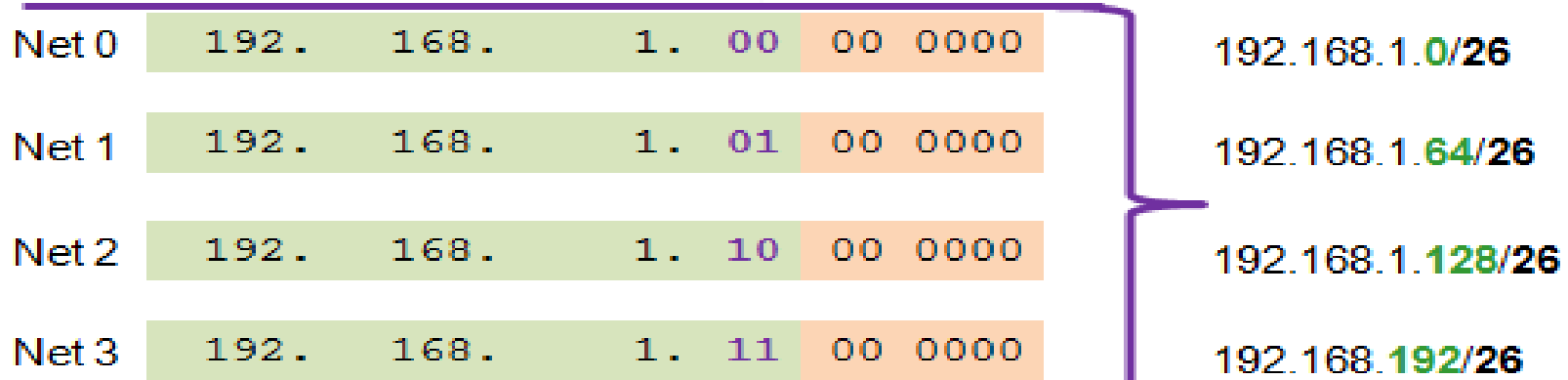
32-26= 6 , 2 power of 6, 64 host on each subnet

Borrowing 2 Bits



Original	192.	168.	1.	00	00	0000
Mask	255.	255.	255.	00	00	0000

Borrowing 2 bits creates 4 subnets:



Net 0	192.	168.	1.	00	00	0000	192.168.1.0/26
Net 1	192.	168.	1.	01	00	0000	192.168.1.64/26
Net 2	192.	168.	1.	10	00	0000	192.168.1.128/26
Net 3	192.	168.	1.	11	00	0000	192.168.1.192/26

All 4 subnets use the same mask:

Mask	255.	255.	255.	11	00	0000	Mask: 255.255.255.192
------	------	------	------	----	----	------	-----------------------

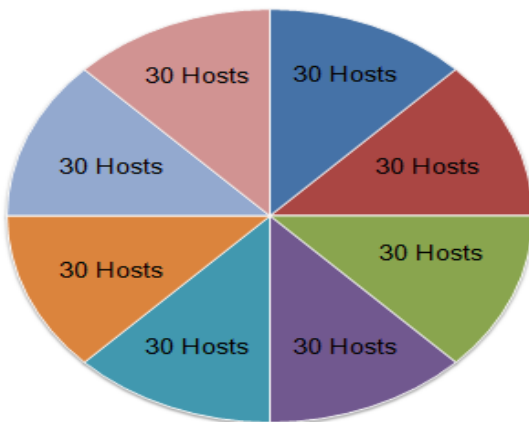
Communication Between Subnets

- A **router** is necessary for devices on different networks and subnets to communicate.
- Each router interface must have an IPv4 host address that belongs to the network or subnet that the router interface is connected to.
- Devices on a network and subnet use the router interface attached to their LAN as their **default gateway**.

TRADITIONAL (FLSM) SUBNETTING WASTES ADDRESSES

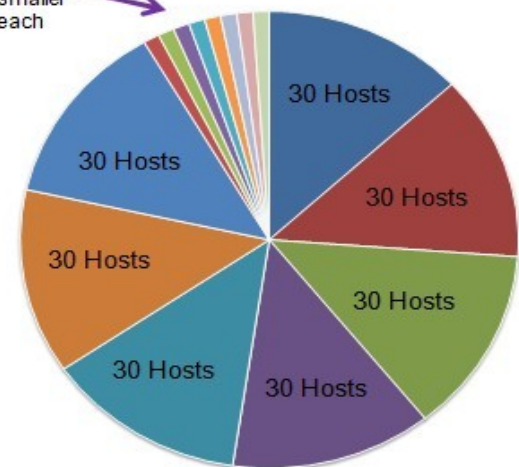
- **Traditional subnetting** - **same** number of addresses is allocated for each subnet.
- Subnets that require fewer addresses have **unused** (**wasted**) addresses. For example, WAN links only need 2 addresses.
- Variable Length Subnet Mask (VLSM) or subnetting a subnet provides more efficient use of addresses.

Traditional Subnetting Creates Equal Sized Subnets



Subnets of Varying Sizes

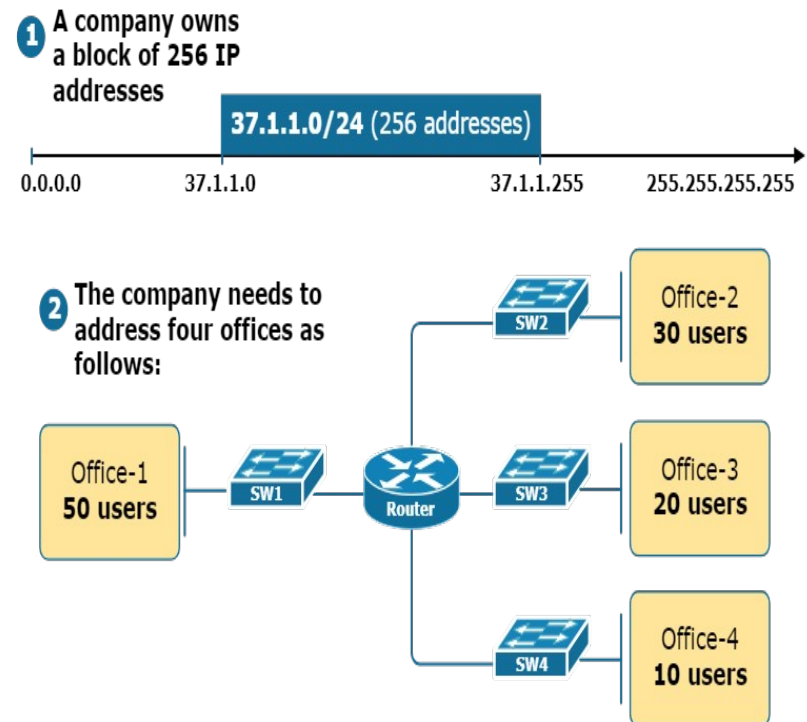
One subnet was further divided to create 8 smaller subnets of 2 hosts each



Variable Length Subnet Masks

Variable length subnet masking (VLSM)

- VLSM allows a network space to be divided in **unequal parts**.
- Allows different masks on the subnets depending on how many bits have been borrowed for a particular subnet.
- Essentially done by subnetting the subnets
- **Network is first subnetted, and then the subnets are subnetted again.**
- Process repeated as necessary to create subnets of various sizes.



Procedure of implementing VLSM – In VLSM, subnets use block size based on requirement so subnetting is required multiple times

- **Suppose** there is an administrator that has **four departments** to manage.
 - These are
 - Sales and purchase department with 120 computers,
 - Development department with 50 computers,
 - Accounts department with 26 computers and
 - Management department with 5 computers.
 - If the administrator has IP 192.168.1.0/24, department wise IPs can be allocated by following these steps:

1.For each segment select the block size that is greater than or equal to the actual requirement which is the sum of host addresses, broadcast addresses and network addresses.

SLASH	NOTATION	HOSTS/SUBNETS
	/24	254
	/25	126
	/26	62
	/27	30
	/28	14
	/29	6
	/30	2

Procedure of implementing VLSM – In VLSM, subnets use block size based on requirement so subnetting is required multiple times

- Arrange all the segments in **descending** order based on the block size that is from highest to lowest requirement.

```
Sales and Purchase: 120  
Development: 50  
Accounts: 26  
Management: 5
```

1. The highest IP available has to be allocated to **highest requirement** so the **sales and purchase** department gets 192.168.1.0/**25** which has 126 valid addresses that can easily be available for 120 hosts. The subnet mask used is 255.255.255.128

2. The next segment requires an IP to handle 50 hosts. The IP subnet with network number 192.168.1.128/**26** is the next highest which can be assigned to 62 hosts thus fulfilling the requirement of development department. The subnet mask used is 255.255.255.192

3. Similarly the next IP subnet 192.168.1.192/**27** can fulfill the requirements of the accounts department as it has 30 valid hosts IP which can be assigned to 26 computers. The mask used is 255.255.255.224

4. The last segment requires 5 valid hosts IP which can be fulfilled by the subnet 192.168.1.224/**29** which has the mask as 255.255.255.248 is chosen as per the requirement.

The End of Tutorials!

Q?

Good Luck!